

RESEARCH ARTICLE

REFRESH—Reversible Embedded Bioprinting for Faster Reinforcement and Structuring of dECM Hydrogels

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ABSTRACT

Fabrication of complex, multi-layered tissue architecture using decellularized extracellular matrix (dECM)-based hydrogel ink is fundamentally limited by the biomaterial's inherent mechanical fragility and slow gelation kinetics, which severely compromise structural fidelity. Herein, we present REFRESH—Reversible Embedded Bioprinting for Faster Reinforcement and Structuring of dECM Hydrogels, a next-generation embedded bioprinting platform designed for high-fidelity fabrication of anatomically relevant tissue constructs using dECM hydrogels. At the core of this system is a custom-engineered polyethylene glycol (PEG)-gelatin microgel suspension bath that performs dual functions: it modulates the bath's rheological behavior, enhancing yield stress, shear-thinning, and self-healing properties necessary for precise deposition, and actively promotes in situ gelation of dECM hydrogel inks via hydrogen bonding and crowding-induced interactions. This enables faster filament stabilization, reduced structural fusion, and improved print fidelity compared to conventional thermal gelation. We recreated the zonal architecture of the trachea by co-printing cartilage- and trachealis muscle-derived dECM's encapsulating primary chondrocytes, fibroblasts, and subsequent epithelialization to form a biomimetic luminal surface. Furthermore, we fabricated functional trifurcated bronchial structures using lung-derived dECM that supported stromal-like mesenchymal behavior with contractile marker expression. The versatility of the REFRESH platform was further validated by the successful printing of a diverse array of tissue-specific dECM hydrogels beyond the airway system.

1 | Introduction

Embedded bioprinting enables the fabrication of soft hydrogel constructs by extruding hydrogel ink into yield-stress suspension baths that provide temporary mechanical stabilization during deposition [1]. This strategy has significantly expanded the printable space of otherwise unprintable materials, particularly low-viscosity hydrogels and biologically derived matrices. However, current suspension baths operate primarily as passive mechanical supports, stabilizing printed filaments without actively influencing hydrogel network assembly [2, 3].

This passive behavior is especially restrictive for decellularized extracellular matrix (dECM) hydrogels, which are valued for

their tissue-specific biochemical complexity and capacity to guide cell behavior, yet exhibit inherently weak pre-gel mechanics and slow gelation kinetics [4]. These properties result in filament relaxation, interlayer fusion, and poor structural fidelity during multilayer printing. Thus, despite their biological advantages, the processing of dECM hydrogels remains a major barrier to their use as functional bioinks [5, 6].

A range of suspension bath strategies has been reported to mitigate filament collapse during printing [7]. These include Carbopol-based yield-stress fluids, alginate and agarose granular slurries, Pluronic-based systems, and gelatin microgel baths, with the latter forming the basis of FRESH-type printing [1–3, 8–18]. These systems provide cytocompatibility, optical transparency,

and thermoreversibility, enabling the printing of collagen- and ECM-based hydrogels that would otherwise deform in air. However, existing baths largely function as mechanically supportive yet chemically inert media, offering little control over the slow and heterogeneous gelation behavior of dECM hydrogels [4]. As a result, print fidelity and structural stability remain tightly coupled to the intrinsic gelation kinetics of the hydrogel ink rather than the capability of the suspension bath itself [19–21]. This situation exposes a fundamental limitation in the current embedded bioprinting paradigm: the suspension bath defines the performance ceiling when printing soft ECM-derived hydrogels, yet it remains passive [19, 22]. A key unresolved question is therefore whether the suspension bath can be engineered to function not only as a support medium, but also as an active regulator that reinforces and accelerates dECM network formation during printing.

Here, we address this unmet need by introducing REFRESH (Reversible Embedded Bioprinting for Faster Reinforcement and Structuring of dECM Hydrogels) — a dual-function PEG–gelatin microgel suspension bath specifically designed to transform the role of the suspension bath from passive stabilization to active hydrogel structuring [23]. REFRESH provides reversible, yield-stress-mediated mechanical support during deposition, while simultaneously accelerating and reinforcing dECM network formation through tailored physicochemical interactions within the bath. The REFRESH bath exhibits thermoresponsive, self-healing behavior, enabling solid-like mechanical support during printing and gentle liquefaction under physiological conditions for non-destructive construct retrieval. This dual functionality preserves high-fidelity spatial architecture even in soft, low-viscosity dECM hydrogels, overcoming long-standing challenges of delayed gelation, weak pre-gel mechanics, and insufficient filament stability.

We demonstrate that REFRESH enables the high-resolution fabrication of structurally organized multilayered airway constructs that recapitulate the biochemical and architectural heterogeneity of the native tracheobronchial system. The versatility of the PEG–gelatin bath supports its application across a wide range of dECM hydrogel inks. Taken together, this work establishes REFRESH as a conceptual and technological advance in embedded bioprinting — redefining the suspension bath as an active reinforcement and structuring system rather than a passive matrix — and providing a robust pathway toward anatomically precise and biologically relevant soft-tissue constructs.

2 | Result and Discussion

In this study, we developed and systematically evaluated REFRESH, a PEG–gelatin microgel-based–suspension bath engineered to address the long-standing challenges of embedded bioprinting with soft, slow-gelling dECM hydrogel inks. Through controlled microgel fabrication, PEG-mediated tuning of physicochemical properties, and rheological optimization, REFRESH functions as a dynamic, yield-stress support medium that stabilizes printed filaments while preserving the integrity of hydrogel inks and cell function. We first characterize the bath architecture, optical clarity, and thermoresponsive behavior to establish its suitability for embedded bioprinting. We then correlate the rheological and printability properties of tissue-specific dECM

hydrogel inks with their performance in REFRESH, providing a mechanistic framework linking material behavior to printing outcomes. The results presented in this section evaluate the capability of REFRESH to function as an active, tunable embedded bioprinting platform for structurally precise and biologically relevant soft-tissue fabrication.

Throughout the manuscript, suspension bath formulations are denoted using a shorthand notation based on their PEG and gelatin contents (w/v). Specifically, 5% PEG–5% gelatin, 5% PEG–10% gelatin, and 0% PEG–10% gelatin formulations are referred to as 5P–5G, 5P–10G, and 0P–10G, respectively.

2.1 | Fabrication and Physicochemical Characterization of the PEG–Gelatin Microgel–Based Suspension Bath

The PEG–gelatin microgel–based suspension bath was designed to provide mechanical support, optical clarity, thermal reversibility, and tunable rheology suitable for high-fidelity embedded bioprinting (Figure 1). Microgels were generated using a water-in-oil emulsification process as described in the experimental section (Figure 1a). This strategy produced reproducible, densely packed microgel slurries in which both composition and packing density could be independently controlled.

Particle-size analysis revealed mean microgel diameters of $20.21 \pm 8.49 \mu\text{m}$ (5P–5G) and $23.69 \pm 9.54 \mu\text{m}$ (5P–10G), with size distributions of 5–42 μm and 8–59 μm , respectively (Figure 1b). In contrast, the gelatin-only control (0P–10G) exhibited larger, more polydisperse particles ($35.75 \pm 14.66 \mu\text{m}$; 13–95 μm). This shift toward smaller and more uniform particles with PEG incorporation is consistent with previous reports showing that PEG modulates droplet breakup and stabilization during emulsification, resulting in narrower size distributions [24]. Microgel sizes within the ~10–50 μm range have also been associated with favorable yield-stress behavior for embedded printing, supporting the functional relevance of these formulations [18].

Optical microscopy confirmed a densely packed granular microstructure in all PEG–gelatin formulations, while SEM imaging revealed discrete spherical microgels embedded within a continuous matrix (Figure 1c). The 5P–5G formulation displayed the most uniform and tightly packed architecture, whereas increasing the gelatin content (5P–10G) produced larger and more polydisperse domains without disrupting network continuity. In contrast, the gelatin-only bath contained larger, sparsely distributed particles with reduced inter-particle contact, consistent with a weaker jammed network. Similar relationships between particle packing density and yield-stress response have been reported in other microgel-based suspension baths, where tighter packing enhances filament stabilization and self-healing during printing [18]. Macroscopically, all formulations appeared solid-like at 25°C, consistent with a jammed microgel network (Figure 1d). At 37°C, the PEG–gelatin baths softened while remaining cohesive, enabling gentle post-printing removal, whereas the gelatin-only bath underwent marked liquefaction. This PEG-mediated thermal stabilization aligns with previous observations that PEG–gelatin interactions, including hydrogen bonding and macromolecular crowding, can increase gelatin

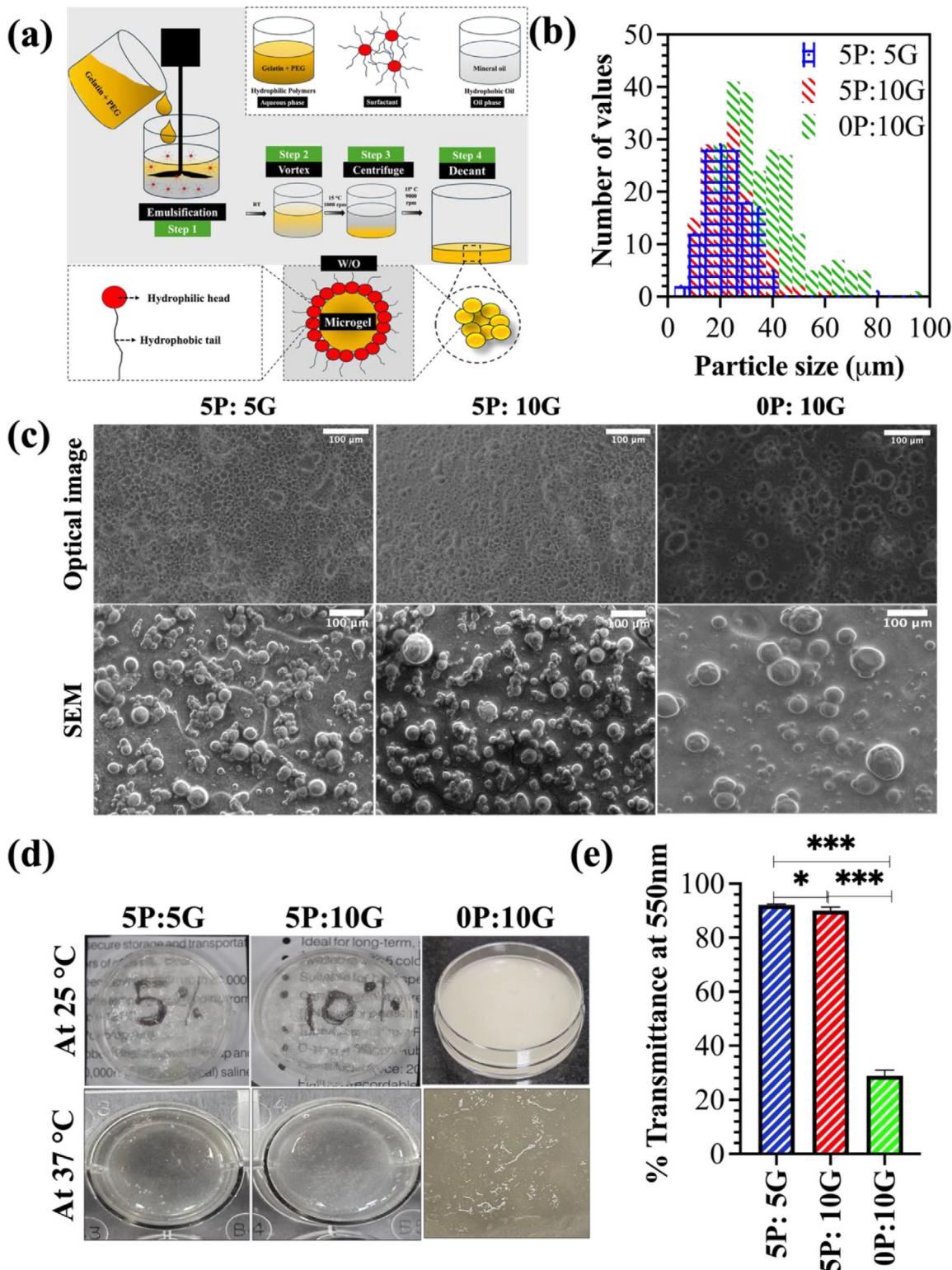


FIGURE 1 | Fabrication, physicochemical characterization, and macroscopic properties of the PEG–gelatin microgel-based REFRESH suspension bath, where formulations containing PEG (5P:5G and 5P:10G) are test groups and the gelatin-only formulation (0P:10G) was used as the control. (a) Schematic illustration of PEG–gelatin microgel fabrication via a water-in-oil (W/O) emulsification process. An aqueous phase containing gelatin and PEG is dispersed into an oil phase with surfactant, followed by vortexing, centrifugation, and decantation to obtain densely packed microgels. (b) Particle size distribution of PEG–gelatin microgels under different formulation conditions, demonstrating tunable microgel size suitable for embedded bioprinting applications. (c) Brightfield and scanning electron microscopy (SEM) images of PEG–gelatin microgels at different compositions, showing uniform spherical morphology and interconnected packing; scale bars: 100 μm . (d) Optical images of 5P-5G, 5P-10G, and 0P-10G baths at 25 °C and 37 °C, respectively. (e) Quantitative comparison of the % transmittance at 550 nm of the formulated suspension baths. Statistical significance was determined by an unpaired two-tailed Student's *t* test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant).

network stability and extend the operational temperature window [25]. Optical measurements further showed >90% transmittance at 550 nm for PEG-containing baths compared with <30% for the gelatin-only control (Figure 1e). Similar increases in optical clarity have been reported in microgel systems with narrower particle-size distributions, where reduced light scattering improves visualization during printing [26].

Our observations are consistent with prior reports indicating that polymer-modified microgels can be engineered to display tunable yield-stress behavior, thermal responsiveness, and optical transparency suitable for embedded bioprinting [24, 26, 27]. Moreover, PEG incorporation into gelatin microgels enables coordinated control of particle size, packing density, clarity, and suspension stability within a single biocompatible bath tailored for delicate dECM inks [28, 29]. For context, Figure S1 illustrates the native morphology of unmodified commercial gelatin prepared as a 10% (w/v) melt before microgel formation.

2.2 | Rheological Characterization of PEG–Gelatin Microgel Suspension Baths for Embedded Bioprinting

The rheological properties of the suspension bath dictate its ability to locally fluidize during nozzle movement while rapidly recovering to a solid-like state for filament stabilization. PEG–gelatin baths (5P–5G and 5P–10G) were therefore compared with a gelatin-only control (0P–10G) to evaluate shear-dependent flow, viscoelasticity, thermal response, and thixotropic recovery (Figure 2).

All formulations displayed pronounced shear-thinning behavior, with viscosity decreasing by several orders of magnitude as shear rate increased (Figure 2a). For 5P–10G, viscosity declined from $\sim 1.4 \times 10^5$ Pa·s at 0.1 s^{-1} to ~ 203 Pa·s at 1000 s^{-1} (~ 690 fold), while reductions for 5P–5G and 0P–10G were ~ 320 fold and ~ 570 fold, respectively. PEG-containing baths exhibited higher viscosity at low shear than the gelatin control, indicating a more densely connected microgel network. Similar strong shear-thinning behavior has been reported in other microgel-based suspension baths, where high zero-shear viscosity supports filament confinement while shear-induced thinning lowers nozzle resistance [6].

A steady shear rate sweep (0.1 – 1000 s^{-1}) was performed, and the yield stress values were obtained directly from the rheometer software based on the viscosity–shear stress profile using the instrument’s built-in algorithm. Shear-stress–shear-rate relationships confirmed yield-stress-dominated flow in all systems. The reported value corresponds to the stress at the onset of flow, defined as the transition from the low-shear plateau region to the shear-thinning regime (without additional manual calculation) with 5P–10G exhibiting the highest yield stress (~ 203 Pa), followed by 5P–5G (~ 129 Pa), while the gelatin-only bath showed a substantially lower value (Figure 2b). A finite yield stress enables the bath to behave as a solid at rest and fluidize only when the applied stress exceeds the threshold, preventing filament sagging and interlayer fusion. This trend is consistent with prior observations that increasing particle packing and interparticle cohesion elevate yield stress in microgel suspensions [6, 30].

Oscillatory strain-sweep testing demonstrated elastic dominance ($G' > G''$) at low strain across all formulations (Figure 2c). Within this regime, the storage modulus was highest for 5P–10G (~ 900 Pa), compared with ~ 300 Pa for 5P–5G and ~ 350 Pa for 0P–10G, and PEG–gelatin baths exhibited a later G' – G'' crossover, indicating greater strain tolerance. These findings align with previous reports that polymer-modified microgels display enhanced stiffness and resistance to structural breakdown, contributing to improved positional stability during embedded printing [31, 32].

Frequency-sweep analysis further demonstrated sustained elastic dominance across the range 0.1 – 100 Hz, with all formulations maintaining G' greater than G'' throughout the tested frequencies (Figure 2d). The magnitude of G' increased with formulation concentration, indicating enhanced network stiffness at higher PEG–gelatin contents. This response corresponds with prior studies identifying predominantly elastic behavior across multiple deformation timescales as critical for limiting creep and distortion in suspension baths [33].

Time-temperature rheology revealed progressive softening of the gelatin-only control near physiological temperature (Figure 2e). Time-temperature rheology revealed distinct thermal stability profiles among the formulations. The gelatin-only system (0P–10G) exhibited high initial stiffness but underwent rapid modulus decay upon approaching 37°C , indicating thermally driven network collapse. The low-gelatin PEG system (5P–5G) displayed significantly lower initial modulus and the fastest structural softening, reflecting insufficient network density to resist thermal disruption. In contrast, 5P–10G maintained the highest modulus retention during heating (~ 550 Pa) and showed delayed transition at physiological temperature, demonstrating enhanced thermo-mechanical stability. This superior stability translated directly to controlled bath leaching behavior, whereas 5P–5G and 0P–10G liquefied more rapidly. These results identify 5P–10G as a well-balanced formulation, providing enhanced mechanical robustness during printing while enabling controlled thermal leaching [30, 34].

Cyclic shear testing demonstrated rapid and reversible thixotropic recovery in PEG–gelatin baths (Figure 2f,g). Thixotropic behavior was evaluated by alternating shear rates of 0.1 s^{-1} and 100 s^{-1} , where 0.1 s^{-1} represents near-rest conditions following filament deposition and 100 s^{-1} approximates localized shear experienced during nozzle movement and extrusion. The 5P–10G formulation retained $\sim 85.6\%$ of its initial viscosity after the first cycle and $\sim 69.7\%$ after the fourth, whereas the gelatin-only bath exhibited severe irreversible breakdown. Comparable recovery behavior has been described in other polymer-modified microgel systems where reversible interparticle interactions support repeated flow-reformation cycles during nozzle traversal [33]. A schematic illustrating nozzle–bath interactions during embedded bioprinting, including localized shear thinning, transient bath distortion, and rapid self-healing that preserves filament geometry, is also provided in Figure S2.

These findings demonstrate that mechanical performance is primarily governed by gelatin concentration, while PEG incorporation enhances network stability when combined with sufficient gelatin content. This synergistic interaction increases yield stress, stiffness, thermal stability, and thixotropic recovery while pre-

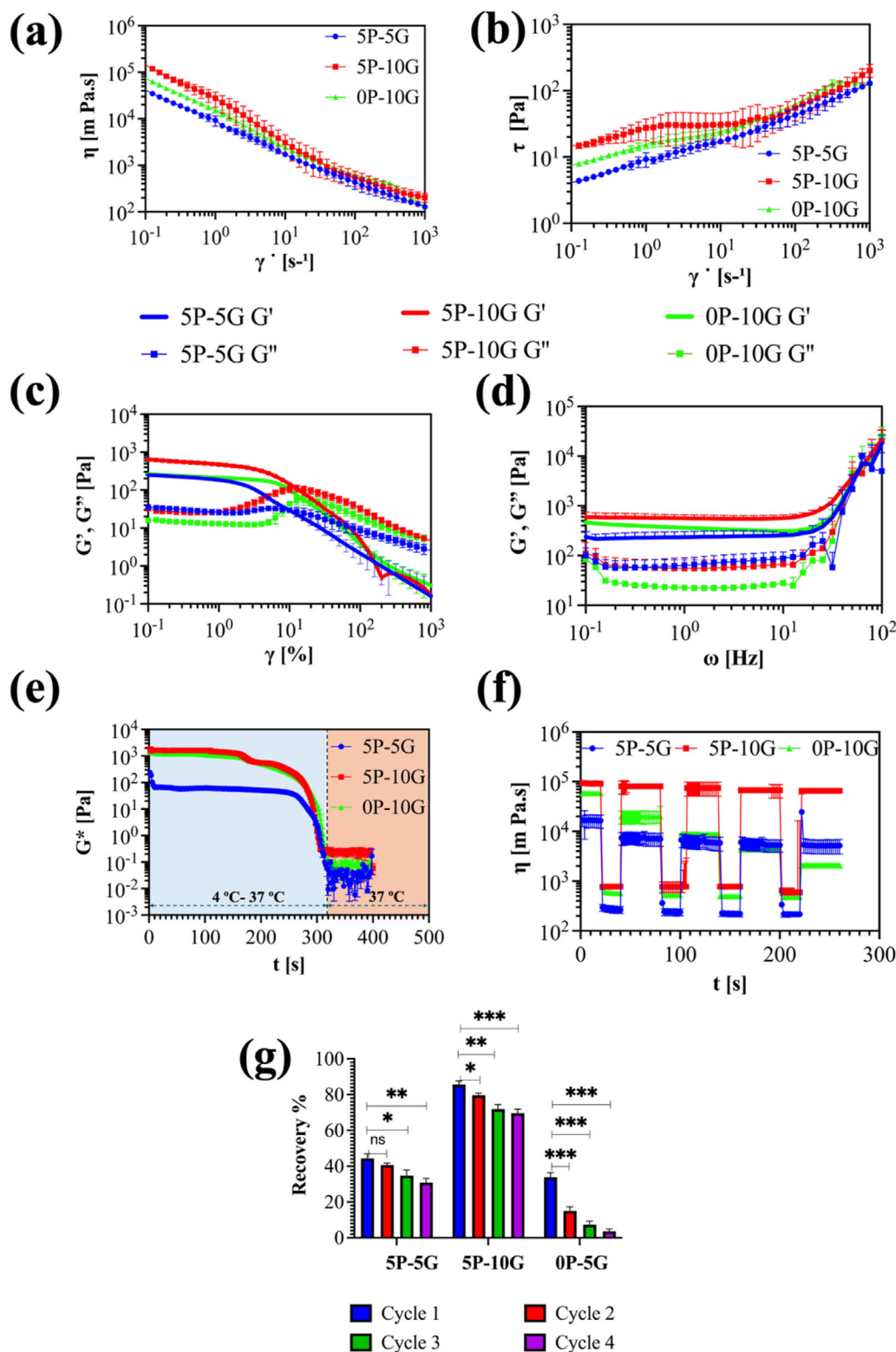


FIGURE 2 | Rheological characterization of the formulated suspension baths (5P-5G and 5P-10G) compared with the control gelatin-only microgel system (0P-10G). (a) Viscosity as a function of shear rate demonstrates pronounced shear-thinning behavior across all formulations. (b) Shear stress vs shear rate curves indicating yield-stress-dominated flow behavior. (c) Storage modulus (G') and loss modulus (G'') as a function of strain, defining the linear viscoelastic region (LVER) and yielding point. (d) Frequency sweep showing the dominance of elastic behavior ($G' > G''$) over the tested frequency range, confirming solid-like characteristics at rest. (e) Time-temperature sweep highlighting the thermal stability and temperature-driven modulation of viscoelastic behavior in the suspension baths. (f) Cyclic high-low shear recovery tests demonstrating pronounced thixotropic behavior, with reproducible structural breakdown under high shear and rapid self-healing upon shear removal over multiple cycles. (g) Quantitative comparison of recovery percentage after shear, highlighting the superior thixotropic recovery and structural resilience of the formulated suspension baths relative to the control. Statistical significance was determined by an unpaired two-tailed Student's t -test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant).

serving strong shear-thinning behavior. In combination with prior work on microgel-based suspension baths, these results establish PEG-stabilized gelatin microgels as a tunable, elastically governed support medium for embedded bioprinting [28, 29, 34]. Among the tested systems, the 5P–10G formulation provided the most favorable mechanical balance for high-fidelity printing of soft, slow-gelling dECM inks.

Further, to quantitatively describe the non-Newtonian behavior of the suspension baths, experimental data were fitted using both Power-law and Herschel–Bulkley (HB) models. The viscosity–shear rate profiles fitted using the Power-law model confirm pronounced shear-thinning behavior across all formulations ($n < 1$) (Figure S3a). The PEG-containing baths exhibited stronger shear sensitivity compared to the gelatin-only system. Shear stress–shear rate data fitted using the Herschel–Bulkley model further improved the description of flow behavior, reflecting the yield-stress-dominated nature of the microgel suspensions (Figure S3b). The inclusion of a yield stress term (τ_0) provided superior correlation with experimental trends.

The extracted Power-law parameters demonstrate reduced flow behavior indices (n) and elevated consistency indices (K), indicating enhanced shear-thinning combined with greater low-shear resistance (Figure S3c). Similarly, analysis of the Herschel–Bulkley parameters reveals that the 5P–10G formulation exhibits the highest yield stress (τ_0) and consistency index (K), while maintaining strong shear-thinning behavior ($n < 1$) (Figure S3d).

These quantitative trends align with oscillatory rheology results and explain the superior filament confinement and dimensional stability observed during embedded printing with the suspension bath. For clarity, the optimized 5P–10G PEG–gelatin suspension bath will be referred to as the REFRESH bath throughout the manuscript.

2.3 | Preparation and Characterization of Tracheal Tissue-derived dECM Hydrogels

Caprine tracheal cartilage and trachealis muscle were successfully processed into tissue-specific decellularized extracellular matrix (dECM) using a non-detergent, salt-based decellularization strategy followed by lyophilization and pepsin-mediated solubilization (Figure S4a). This workflow was designed to achieve effective cellular removal while minimizing disruption to the native extracellular matrix architecture, which is essential for preserving tissue-specific biochemical and mechanical cues.

Quantitative biochemical analysis confirmed effective decellularization of both cartilage-derived (dCM) and muscle-derived (dMM) extracellular matrices while preserving their tissue-specific biochemical profiles (Figure 3a). Residual DNA levels in both hydrogels were minimal and remained well below the commonly accepted threshold of 50 ng/mg for decellularized biomaterials, indicating efficient cellular removal and a low risk of immune response [4]. These values are comparable to previous reports for dECM materials used in regenerative applications, supporting the effectiveness of the salt-based decellularization strategy employed here [35]. Non-detergent decellularization preserved a substantial fraction of native ECM components, with

dMM retaining ~60–70% and dCM retaining ~40–50% of GAG and collagen contents ($***p < 0.001$ vs. native) (Figure 3b,c). The substantial preservation of ECM components indicates that the non-detergent salt-based protocol successfully maintains the native biochemical milieu required for functional tissue engineering.

Selective retention of key ECM components is critical, as matrix composition directly regulates hydrogel hydration, mechanical behavior, and cell-mediated signaling. The compositional differences observed between dCM and dMM provide a mechanistic basis for the distinct viscoelastic and rheological responses exhibited by the two dECM hydrogels, consistent with prior studies demonstrating that native tissue composition strongly influences downstream hydrogel performance [33]. Histological evaluation further corroborated the biochemical findings while demonstrating preservation of native ECM architecture (Figure S4b(i–viii)). DAPI staining showed near-complete removal of nuclear material in decellularized matrices relative to native tissue, confirming the absence of DNA-containing cellular remnants. Complementary H&E staining revealed no intact cells while maintaining key structural features, including the lacunar morphology of cartilage and the fibrous organization of trachealis muscle. These results indicate that the salt-based protocol removed cellular content while minimizing disruption to hierarchical ECM structure. Similar preservation of matrix architecture following gentle decellularization has been reported in other airway-relevant dECM systems and is considered advantageous for maintaining biological function [36].

Taken together, the biochemical and histological data demonstrate that the decellularization strategy successfully removed cellular material while retaining both the composition and microstructural organization of tissue-specific ECM. Preservation of these hierarchical features is particularly important, as they influence enzymatic solubilization, hydrogel self-assembly, and subsequent cell–matrix interactions. These findings reinforce growing evidence that decellularization is not solely a preparative step but a key determinant of the biological instructiveness and mechanical behavior of resulting dECM hydrogels [4]. Accordingly, the well-preserved dCM and dMM matrices produced here provide a biologically relevant and mechanically tunable material platform for airway tissue engineering and biofabrication.

Both cartilage-derived (dCM) and muscle-derived (dMM) ECM solutions underwent sol–gel transition at 37°C, forming stable, self-supporting hydrogels (Figure S4c(i–ii)). This confirms that solubilized ECM components retain sufficient intermolecular interactions to reassemble into physically cross-linked networks. Although gelation occurred under identical conditions, visible differences in gel firmness and opacity suggested that tissue-specific composition influences network density and cross-linking efficiency, consistent with prior dECM reports [4]. Swelling analysis showed controlled hydration in both hydrogels without dissolution and structural failure, indicating robust network formation (Figure S4d). dCM displayed a higher swelling ratio than dMM, consistent with its greater glycosaminoglycan-mediated water-binding capacity, whereas the collagen-rich dMM hydrogel showed more restrained swelling. These findings align with literature demonstrating composition-dependent hydration behavior in ECM-derived hydrogels [4]. These results

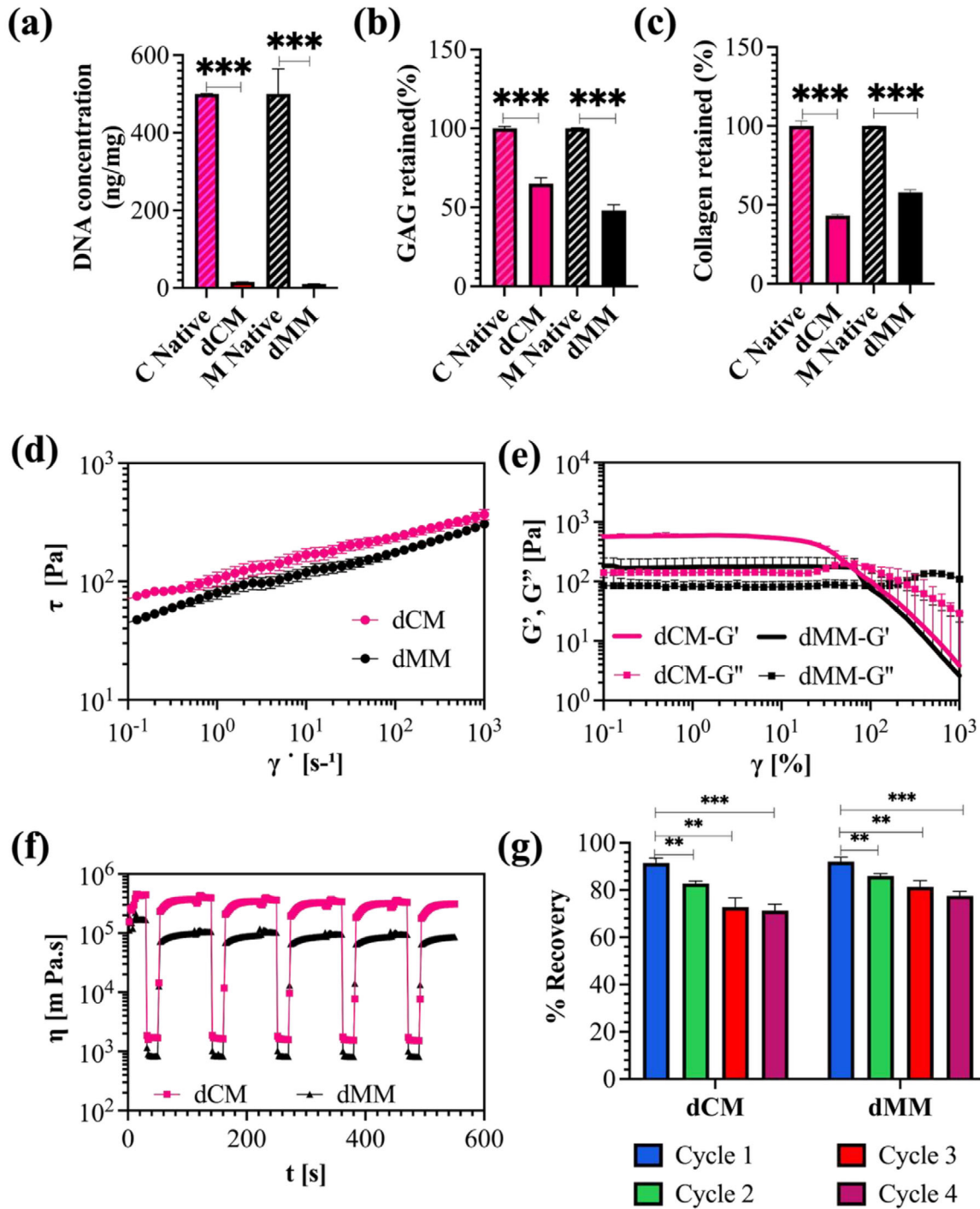


FIGURE 3 | Biochemical composition and rheological characterization of tissue-specific dECM hydrogels. (a-c) Biochemical characterization of decellularized cartilage matrix (dCM) and decellularized trachealis muscle matrix (dMM) showing residual DNA content, glycosaminoglycan (GAG) content, and collagen content, respectively. Statistical significance determined by unpaired two-tailed Student's *t* test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant). (d) Shear stress vs shear rate curves revealing yield-stress-dominated flow behavior, confirming the presence of a percolated ECM network capable of maintaining structural integrity at rest while flowing under applied shear. (e) Oscillatory strain sweep showing storage modulus (G') and loss modulus (G'') as a function of strain, defining the linear viscoelastic region (LVR) and yielding behavior of the dECM hydrogels. (f) Cyclic high-low shear thixotropy tests illustrating reversible structural breakdown under high shear followed by rapid recovery upon shear removal, demonstrating robust thixotropic and self-healing behavior. (g) Quantitative analysis of recovery percentage after repeated shear cycles, highlighting the efficient structural recovery and rheological resilience of both dCM and dMM hydrogels. Statistical significance was determined by an unpaired two-tailed Student's *t* test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant).

confirm that dCM and dMM form stable, physically cross-linked hydrogels under physiological conditions while preserving tissue-specific differences in hydration and network structure. Such intrinsic variation is advantageous for engineering tracheal constructs that require spatially distinct mechanical and transport properties.

2.4 | Rheological Characterization of Tissue-Specific dECM Hydrogels

Rheological performance is a critical determinant of dECM hydrogel suitability for extrusion-based and embedded bioprinting, as shear-dependent viscosity, viscoelastic moduli, and thermally triggered gelation dictate nozzle flow behavior, post-extrusion filament stabilization, and subsequent mechanical integrity [33]. Because these properties arise directly from native ECM composition—particularly collagen and glycosaminoglycan (GAG) content—cartilage-derived (dCM) and muscle-derived (dMM) dECM hydrogels were systematically evaluated under printing-relevant conditions to correlate biochemical identity with mechanical behavior (Figure 3d–g).

Both hydrogels exhibited yield-stress-dominated flow (Figure 3d). At low shear ($\sim 0.01 \text{ s}^{-1}$), dCM showed a higher yield stress than dMM ($\sim 70 \text{ Pa}$ vs. $\sim 27 \text{ Pa}$), indicating greater resistance to deformation before flow onset. Even at high shear (1000 s^{-1}), dCM retained slightly higher resistance ($\sim 369 \text{ Pa}$ vs. $\sim 306 \text{ Pa}$). These findings indicate the formation of percolated ECM networks that remain solid-like until a critical stress is exceeded. The higher low-shear resistance of dCM is consistent with its greater GAG content, which enhances hydration-mediated network reinforcement, whereas the collagen-rich dMM matrix permits earlier rearrangement. Yield-stress-governed flow of this type is widely reported for dECM hydrogels and is advantageous for maintaining structural fidelity following deposition [33].

Both dECM hydrogels demonstrated pronounced shear-thinning (Figure S5a). Across 0.1 – 1000 s^{-1} , dCM displayed a ~ 3800 fold viscosity reduction compared with ~ 1600 fold for dMM, confirming stronger rate-dependent thinning in dCM. dCM also maintained higher viscosity at low shear, reflecting a denser and more hydrated polymer network. Such viscosity profiles are characteristic of physically cross-linked ECM hydrogels and support smooth extrusion under shear while preserving high resistance at rest, as reported for other dECM printing systems [33].

Oscillatory strain sweeps showed elastic-dominated behavior at low strain for both hydrogels ($G' > G''$), consistent with solid-like viscoelastic networks (Figure 3e). dCM exhibited higher low-strain stiffness than dMM but also experienced greater modulus reduction at high strain, indicating larger non-linear softening. Both hydrogels displayed a defined linear viscoelastic range followed by $G' - G''$ crossover, marking the onset of yielding. These behaviors align with reports that ECM hydrogels dissipate stress through reversible disruption of weak physical interactions before permanent network breakdown [33].

Dynamic frequency sweeps confirmed sustained elastic dominance across 0.1 – 100 Hz for both dCM and dMM (Figure

S5b). dCM maintained consistently higher absolute moduli, whereas dMM exhibited stronger frequency-dependent stiffening. The relatively weak frequency sensitivity of G' suggests stable network architecture over physiologically relevant deformation timescales, consistent with literature describing dECM hydrogels as mechanically resilient yet dynamic materials [33]. These properties support mechanical stability during handling and bioprinting.

Time-temperature sweeps revealed a clear sol–gel transition for both hydrogels, with rapid G' increase at 37°C followed by plateau stabilization (Figure S5c). dCM gelled slightly earlier and achieved a higher plateau modulus than dMM, reflecting tissue-specific differences in assembly kinetics and final network density. These thermally triggered transitions are consistent with the temperature-dependent reorganization of solubilized ECM reported in other studies [33] and confirm that both materials undergo reproducible gelation under physiologic conditions.

Cyclic Thixotropy shear testing demonstrated reversible structural breakdown followed by rapid recovery in both hydrogels (Figure 3f,g). Thixotropic recovery was assessed by alternating shear rates of 0.1 s^{-1} and 100 s^{-1} , where 0.1 s^{-1} represents near-rest conditions after filament deposition and 100 s^{-1} approximates the elevated shear experienced during extrusion and nozzle movement. Recovery remained above 70% across cycles, with dMM consistently exhibiting greater recovery efficiency ($\sim 92\% \rightarrow \sim 78\%$ across Cycles 1–4) compared with dCM ($\sim 92\% \rightarrow \sim 71\%$). This resilience reflects reversible, non-covalent interactions characteristic of ECM hydrogels and is considered essential for maintaining filament continuity and dimensional fidelity during repeated nozzle traversal, consistent with previous dECM reports [33].

These findings demonstrate that tracheal cartilage- and muscle-derived dECM hydrogels possess yield-stress-governed, shear-thinning, viscoelastic, and thixotropic properties that enable smooth extrusion, rapid stabilization, and mechanical resilience—features regarded as optimal for embedded bioprinting [33]. The higher stiffness and low-shear resistance of dCM derive from its GAG-rich composition, while the greater recovery efficiency of dMM reflects its collagen-dominated fibrillar structure. These distinct yet complementary rheological signatures provide a rational basis for constructing spatially heterogeneous tracheal models that better recapitulate native cartilage–muscle organization and function.

The shear-dependent flow behavior of dCM and dMM hydrogel inks was analyzed using Power-law and Herschel–Bulkley (HB) models (Figure S6a–d). Both hydrogels exhibited pronounced shear-thinning behavior across the tested shear rate range (Figure S6a), confirming their suitability for extrusion-based printing. Notably, dCM displayed consistently higher viscosity than dMM at low shear rates, reflecting its denser ECM composition. Shear stress–shear rate profiles further demonstrated nonlinear flow behavior characteristic of yield-stress fluids (Figure S6b). Power-law fitting revealed a lower flow behavior index for dCM ($n = 0.10$) compared to dMM ($n = 0.176$), indicating stronger shear-thinning in the cartilage-derived matrix (Figure S6c). Correspondingly, the higher consistency index (K) of dCM suggests greater resistance to deformation under low-shear conditions.

Herschel–Bulkley analysis confirmed the presence of finite yield stress in both inks, with dCM exhibiting a higher τ_0 (4.99 Pa) than dMM (3.06 Pa) (Figure S6d). This combination of elevated yield stress and stronger shear-thinning behavior in dCM explains its improved filament stability after deposition but also its greater extrusion resistance at higher speeds. In contrast, the lower τ_0 and moderate shear-thinning of dMM facilitate smoother extrusion, consistent with the enhanced continuity observed during printing optimization. These rheological differences quantitatively explain the distinct extrusion resistance and filament continuity observed during printing.

2.5 | PEG-Mediated Modulation of dECM Gelation Kinetics and Network Formation

Slow, heterogeneous, and diffusion-limited gelation is a major limitation of dECM hydrogels, particularly under ambient or sub-physiological conditions where fibrillogenesis proceeds inefficiently. In this work, PEG was incorporated into the REFRESH suspension bath not only as a rheological modifier, but as an active physicochemical regulator of dECM assembly via macromolecular crowding (MMC). MMC is increasingly recognized as a key mechanism by which neutral polymers such as PEG exclude solvent volume, increase effective macromolecule concentration, and thereby accelerate collagen and dECM network formation [25]. The mechanistic role of PEG in accelerating dECM gelation is described in Section 4. A schematic illustrating PEG-induced macromolecular crowding (MMC)-mediated enhancement of dECM fibrillogenesis is provided in Figure S7. To isolate the contribution of MMC from native hydrogen-bond-mediated interactions, urea was employed as a hydrogen-bond disruptor [37, 38], providing a chemically perturbed system in which to test whether PEG can still promote ECM assembly.

Time-resolved turbidity assays revealed clear formulation-dependent differences in fibrillogenesis kinetics (Figure 4a). Native dECM displayed a gradual and incomplete turbidity increase, consistent with slow network formation. Incorporation of 5% PEG (REFRESH condition) markedly steepened the turbidity curve and increased the final turbidity plateau, indicating accelerated and more extensive fibrillar assembly. As expected, urea strongly suppressed turbidity development, confirming effective disruption of hydrogen-bond-mediated interactions [37, 38]. Strikingly, PEG continued to accelerate fibrillogenesis even in the presence of urea, with turbidity recovery approaching levels seen in untreated PEG-containing dECM [39]. Quantitatively, turbidity increased ~12 fold in native dECM, ~64 fold in dECM+PEG, ~13 fold in dECM+urea, and ~47 fold in dECM+urea+PEG relative to baseline. These findings strongly support macromolecular crowding as the dominant mechanism by which PEG accelerates gelation, with hydrogen bonding acting as a secondary, synergistic contributor rather than the principal driver. Similar PEG-mediated enhancement of dECM gelation has been reported in cardiac dECM systems, where excluded-volume effects reduced gelation time and improved network uniformity without chemical crosslinking [39].

Consistent with these observations, PEG-containing formulations exhibited higher maximum turbidity ($A_{\max}$), shorter gela-

tion half-times ($t_{1/2}$), and increased overall gelation activity (AUC) compared to native dECM, whereas urea treatment suppressed these parameters (Figure 4b–d). These effects parallel previous reports demonstrating that macromolecular crowding accelerates fibrillogenesis by increasing the effective local concentration of ECM precursors and lowering kinetic barriers to nucleation [39]. Complementary time–temperature rheology further confirmed that PEG shifted the sol–gel transition to earlier time points during warming from 4°C to 37°C, while preserving the inherent thermoresponsive nature of dECM assembly (Figure 4e). Scanning electron microscopy provided structural corroboration for these kinetic findings (Figure 4f). Native dECM assembled into a loose and heterogeneous fibrillar network with large voids, consistent with diffusion-limited growth (Figure 4f(ii)) [40]. In contrast, PEG-exposed dECM displayed a denser, more continuous fibrillar architecture with reduced pore size and enhanced lateral aggregation (Figure 4f(i)). This microstructural refinement is characteristic of MMC-driven systems, where crowding promotes efficient nucleation and coordinated fibril–fibril association [41, 42].

Together, these data demonstrate that PEG-induced macromolecular crowding is the primary driver of accelerated and reinforced dECM fibrillogenesis, while hydrogen-bond interactions provide additional but secondary stabilization. By enhancing the rate, extent, and spatial coordination of ECM assembly—even under conditions in which hydrogen bonding is partially disrupted—PEG effectively extends the usable gelation window for embedded bioprinting, stabilizes extruded filaments, and improves construct fidelity. Previous embedded-printing studies employing gelatin, Carbopol, or agarose suspension baths improved filament retention but largely acted as passive mechanical stabilizers and did not regulate the inherently slow gelation of dECM [1, 10, 13, 43]. Moreover, gelatin-only systems often exhibit low yield stress and slow recovery, while Carbopol and agarose lack thermal reversibility and may complicate construct retrieval [16]. In contrast, the PEG–gelatin microgel bath developed here provides jammed-particle mechanical support together with active kinetic regulation of dECM assembly, offering a unique combination of tunability, optical clarity, cytocompatibility, and thermal reversibility ideally suited for high-fidelity embedded bioprinting. This dual-function design establishes REFRESH as a distinct advance over prior suspension baths by coupling physical stabilization with programmable control over dECM network formation, thereby enabling reliable printing of biologically specialized, mechanically fragile matrices.

2.6 | Speed-Dependent Print Fidelity on Flat Substrates and in Embedded 3D Printing

To determine how rheological behavior translates into printing performance, filament formation was systematically evaluated under both conventional surface printing and embedded printing conditions. In flat-surface printing, the influence of translation speed on filament morphology and grid fidelity was examined using cartilage-derived (dCM) and muscle-derived (dMM) dECM hydrogel inks to identify a stable deposition window. In parallel, embedded printing was assessed to determine how suspension bath mechanics influence filament confinement and dimensional stability. Three bath formulations were compared while varying

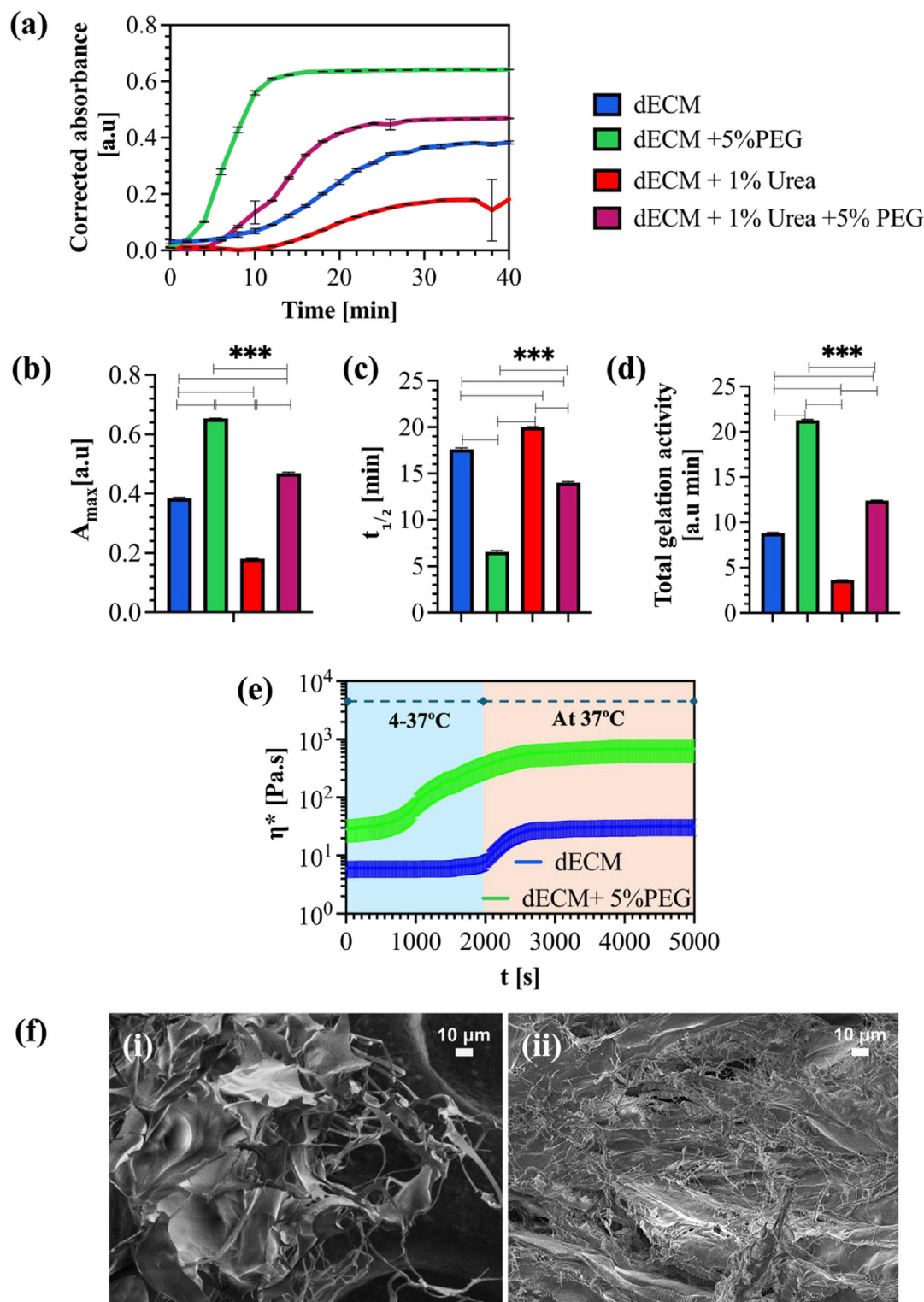


FIGURE 4 | PEG-mediated modulation of dECM gelation kinetics and microstructure. (a) Time-dependent turbidity profiles of dECM hydrogels under different conditions: only dECM, dECM + 5% PEG [REFRESH condition], dECM + 1% urea, and dECM + 1% urea + 5% PEG, demonstrating distinct gelation kinetics and plateau turbidity values. (b-d) Quantitative analysis of turbidity-derived parameters, including (b) maximum turbidity ($A_{\max}$), (c) gelation half-time ($t_{1/2}$), (d) total gelation activity quantified as the area under the turbidity–time curve (AUC) for the indicated dECM formulations highlighting the accelerating effect of PEG and the inhibitory effect of urea on dECM fibrillogenesis. (e) Temperature-dependent gelation behavior of dECM and dECM + 5% PEG [REFRESH condition] during transition from 4°C to 37°C, illustrating enhanced and faster network formation in the presence of PEG. (f) Scanning electron microscopy images of printed dECM constructs fabricated (i) within the PEG-containing REFRESH suspension bath and (ii) within the control bath (0P-10G), revealing enhanced fibrillar density and network interconnectivity upon PEG incorporation. Scale bars: 10 μm . Statistical analysis was performed using the Brown–Forsythe test for homogeneity of variance followed by a one-way ANOVA test ($p^{***} < 0.001$; $n = 3$).

extrusion speed to evaluate their ability to stabilize soft dECM filaments during deposition.

While printing on flat substrates, filament morphology exhibited pronounced speed dependence (Figure 5a). At low speeds ($1\text{--}3\text{ mm s}^{-1}$), excessive volumetric deposition promoted strand spreading, lateral fusion, and pore collapse. This effect was more pronounced for dCM, consistent with its higher low-shear viscosity and slower structural relaxation. Optimization of the gelation state further confirmed the importance of balanced viscoelasticity for filament stability (Figure S8a). Under-gelation resulted in excessive spreading, whereas over-gelation led to discontinuous extrusion and structural irregularities. Proper gelation enabled stable filament formation across speeds. Quantitative analysis of vertical strand width and pore area (Figure S8b–d) corroborated the speed-dependent trends observed in Figure 5, with improved geometric resolution at intermediate-to-high speeds. Notably, formulation-dependent differences between dCM and dMM were preserved across metrics, consistent with their distinct rheological profiles. Increasing translation speed progressively reduced spreading and improved grid definition. Quantitative analysis confirmed a monotonic decrease in strand width with increasing speed for both inks (Figure 5c). A functional surface-printing window was observed between $3\text{--}9\text{ mm s}^{-1}$, with $\sim 6\text{ mm s}^{-1}$ providing the most consistent balance between deposition stability and geometric fidelity. Nevertheless, unsupported printing remained constrained by post-deposition relaxation, reflecting the inherently low modulus of dECM hydrogels. These findings reflect the differential shear-responsiveness of the two dECM inks and align with prior reports that extrusion speed strongly modulates filament fidelity in soft hydrogel systems [33].

Embedded bioprinting within suspension baths using the dCM hydrogel ink markedly enhanced filament confinement (Figure 5b). Strand width decreased with increasing speed across all bath formulations; however, dimensional control was strongly bath-dependent (Figure 5d). The gelatin-only control bath (0P–10G) produced the widest filaments, indicating limited confinement. In contrast, PEG-containing baths generated narrower and more uniform strands across speeds.

Direct comparison between flat-surface and embedded printing revealed substantially reduced spreading within PEG–gelatin baths (Figure 5c,d), consistent with their yield-stress-dominated and rapidly recovering microgel networks. For clarity, the strand width of dCM printed on a flat substrate was directly compared with embedded conditions in Figure 5d, demonstrating significantly greater filament spreading in unsupported printing relative to PEG-containing suspension baths. Among the tested systems, the 5P–10G formulation (REFRESH) provided the most balanced performance, combining strong confinement with stable filament continuity across speeds. These findings demonstrate a mechanistic transition from substrate-governed fidelity during surface printing to bath-governed stabilization in embedded bioprinting, where tunable yield stress suppresses filament distortion and enhances dimensional accuracy.

Representative images illustrating conventional printing in air and embedded printing within the REFRESH bath highlight the enhanced structural stability and geometric preser-

vation achieved under embedded conditions (Figure S9a). The REFRESH suspension bath exhibited the key characteristics required for reliable embedded bioprinting, including strong interparticle cohesion, smooth filament extrusion, high structural fidelity during multilayer stacking, and rapid recovery after shear. These properties collectively supported continuous deposition of soft dECM filaments into stable, three-dimensional constructs. A comparative summary of these suspension-bath behaviors is provided in Table S1. Structural fidelity during embedded printing was next evaluated at the construct scale using the PEG–gelatin REFRESH bath (5P–10G) and a gelatin-only control bath (0P–10G). Representative images of ring-shaped constructs printed in both suspension baths are shown in Figure S10a. Ring-shaped dECM constructs printed in the control bath exhibited geometric distortion and loss of structural definition following bath removal. In addition, the thermoresponsive and residue-free recovery of printed constructs from the bath is detailed in Supporting Information, Section S8 and Figure S12a–d.

2.7 | Cytocompatibility and ECM-Dependent Cellular Organization

To evaluate biological performance within the identified surface printing window, NIH-3T3 fibroblasts were encapsulated in sterile dCM and dMM hydrogel inks and printed at 3, 6, and 9 mm s^{-1} (Figure S11a,b). Live/dead staining at Days 2, 7, and 14 demonstrated sustained high post-printing viability across all conditions (Figure S11c), with quantitative analysis confirming $>90\%$ viability at $3\text{--}6\text{ mm s}^{-1}$ and $>85\%$ at 9 mm s^{-1} over 14 days for both bioinks (Figure S11e,f). These results indicate that shear exposure within the operational range does not compromise cell survival. Despite comparable viability, matrix-specific differences in cytoskeletal organization were evident. Phalloidin staining revealed pronounced F-actin elongation and alignment in dMM constructs, particularly at 6 mm s^{-1} (Figure S11d), consistent with the anisotropic collagen architecture of muscle-derived ECM [44]. In contrast, fibroblasts within dCM constructs exhibited a more rounded morphology with limited alignment, reflecting the isotropic, proteoglycan-rich composition of cartilage-derived ECM [45].

These findings demonstrate that extrusion speed primarily regulates mechanical exposure and geometric fidelity, whereas ECM composition governs post-printing cellular organization. The optimized printing speed ($\sim 6\text{ mm s}^{-1}$) therefore provides a balanced operational point that preserves structural accuracy while maintaining long-term cytocompatibility.

2.8 | Structural Fidelity and Biocompatibility of Embedded Constructs Printed in the Optimized REFRESH Bath

Following comparative evaluation of multiple suspension bath formulations (Section 2.6), the 5P–10G system was identified as the optimal embedded support matrix. We therefore next investigated structural fidelity and biological performance using this optimized formulation, referred to as the REFRESH bath, in embedded 3D printing of dCM- and dMM-based hydrogel inks.

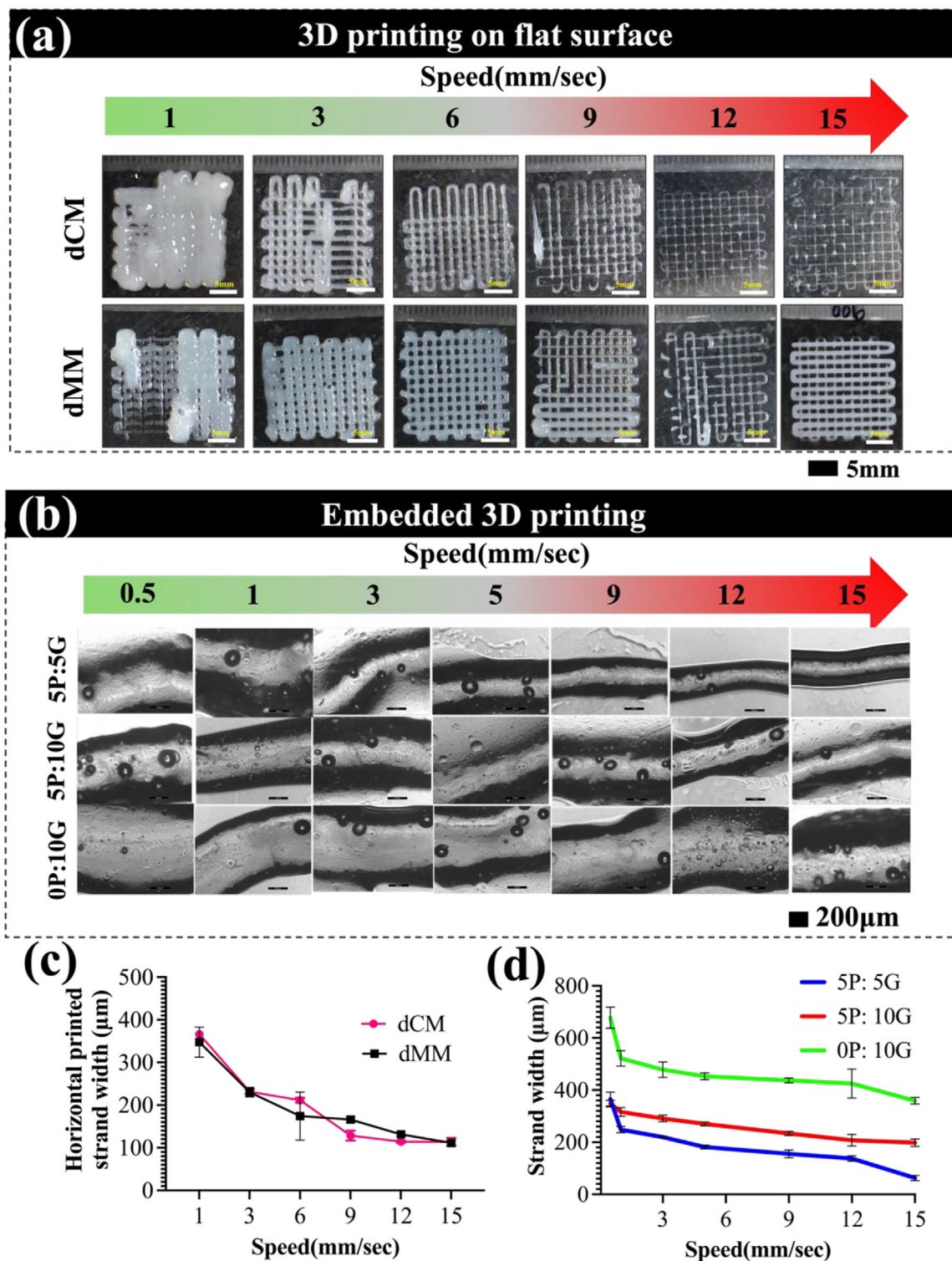


FIGURE 5 | Speed-dependent print fidelity on flat substrates and in embedded 3D printing. (a) Representative images of grid structures printed on a flat surface using two different hydrogel inks across increasing translation speeds (1–15 mm s⁻¹). At low speeds, both hydrogel inks exhibit filament spreading and pore collapse due to over-deposition, while increasing speed improves strand separation and lattice definition, with hydrogel ink-dependent differences in shape retention. Scale bar: 5 mm. (b) Brightfield micrographs of filaments printed using embedded 3D printing in different suspension bath formulations, including the gelatin-only control bath (0P–10G), across increasing printing speeds. Suspension bath composition strongly influences filament confinement, continuity, and surface smoothness, with PEG-containing baths enabling superior lateral stabilization and reduced filament distortion over a broad speed window. Scale bar: 200 µm (c) Quantitative analysis of filament width as a function of printing speed for the two hydrogel inks printed on flat substrates, showing a progressive decrease in filament diameter with increasing speed. (d) Comparative assessment of filament resolution and uniformity across suspension bath formulations, highlighting improved dimensional fidelity relative to the 0P–10G control. Data are presented as mean ± SD (n = 3).

Embedded printing within REFRESH produced continuous, well-confined filaments across the tested speed range (0.5–6 mm s⁻¹) for both dCM and dMM hydrogel inks (Figure 6a). Filament boundaries remained sharply defined, with minimal lateral spreading or interface distortion. Quantitative strand-width analysis demonstrated a controlled decrease in filament diameter with increasing speed for both formulations (Figure 6b), indicating stable shear-induced thinning without structural collapse. The comparable confinement achieved for dCM and dMM—despite their intrinsic compositional and rheological differences—confirms that stabilization is governed primarily by bath mechanics rather than ink-specific behavior. These results are consistent with the yield-stress-dominated and rapidly recovering properties of REFRESH, which transiently fluidizes during nozzle motion and rapidly re-establishes solid-like support following shear removal. Based on these results, an extrusion speed of 1 mm s⁻¹ was selected for subsequent cell-encapsulated printing and culture studies to ensure stable filament formation while minimizing shear exposure to embedded cells.

Cell viability during embedded bioprinting was next evaluated using NIH-3T3 fibroblasts encapsulated in dCM and dMM bioinks. Quantitative viability analysis demonstrated high survival rates at both early (Day 1) and later (Day 14) time points (Figure 6c), indicating that extrusion and bath-mediated confinement did not impose detrimental mechanical stress. Live/dead fluorescence imaging confirmed uniform cellular distribution and sustained viability within printed filaments for dCM and dMM bioinks, respectively (Figure 6d,f). F-actin staining further revealed well-organized cytoskeletal structures with clear spreading and alignment (Figure 6e,g), indicative of active cell–matrix interaction within the stabilized 3D environment.

To assess construct-level performance under more demanding geometries, branching bronchial structures and free-form tongue-like architectures were printed in both REFRESH and the control bath. Constructs produced in REFRESH preserved their intended 3D form and could be handled without deformation, whereas those fabricated in the gelatin-only bath displayed pronounced geometric distortion and reduced handling stability (Figure 6h). These qualitative observations were supported by quantitative mechanical testing. Under tensile loading, REFRESH-printed constructs exhibited significantly higher tensile modulus, ultimate tensile strength, and strain-to-failure compared with control constructs (Figure 6i–k). Compressive testing likewise demonstrated higher compressive modulus and failure stress (Figure 6l–n), and dynamic mechanical analysis revealed consistently greater stiffness across the tested frequency range (Figure 6o). These data indicate that constructs produced in the REFRESH bath possess enhanced resistance to deformation under both static and oscillatory loading. The improved mechanical performance can be attributed to bath-mediated confinement, which promotes filament continuity, suppresses spreading, and enhances interlayer integration during deposition. These effects are consistent with prior work showing that yield-stress baths improve dimensional fidelity and reduce structural collapse in soft hydrogel systems [6]. However, the PEG-reinforced REFRESH bath appears to provide an additional advantage over gelatin-only systems, which exhibit lower yield stress and slower recovery and therefore permit filament distortion and interfacial defects that compromise construct integrity [32].

These findings establish that REFRESH-based embedded bioprinting enables fabrication of free-standing dECM constructs with superior geometric fidelity and mechanically reinforced architectures, whereas comparable structures printed in conventional gelatin-only suspension baths display reduced structural and mechanical stability.

2.9 | Multimaterial Embedded Bioprinting of a Multilayered Construct Using Tissue-Specific Bioinks

Leveraging a dual-cartridge extrusion strategy, cartilage-derived (dCM) and muscle-derived (dMM) decellularized matrix hydrogels were consecutively printed to generate a multilayered cylindrical construct within the PEG–gelatin REFRESH suspension bath (Figure 7a). The yield-stress behavior of the bath enabled precise filament confinement during deposition, followed by thermal gelation and reversible bath leaching to yield a free-standing construct that retained its intended geometry. Optimization of the multimaterial printing process is shown in Figure S13a–c. Multimaterial embedded bioprinting has previously been explored using combinations of synthetic and natural hydrogels to recreate tissue heterogeneity or interfacial regions, such as cardiac and osteochondral constructs. However, most prior studies relied on non-tissue-specific matrices or limited dECM combinations [5, 6, 8, 31, 32, 46].

Here we report a distinct advance by integrating two highly tissue-specific dECM hydrogels of caprine origin — dCM and dMM — within a single embedded-printed construct, inspired by the alternating cartilage–muscle organization of the trachea [47]. The printed structures maintained cylindrical fidelity following bath removal, exhibiting clearly defined, continuous layers in both side-view and cross-sectional imaging (Figure 7b–d). This outcome reflects the combined contribution of the rheologically tuned dECM inks and the stabilizing REFRESH bath, whose finite yield stress provided essential mechanical support during deposition and gelation. However, unlike earlier approaches that primarily employed non-tissue-specific or single-component bioinks, the REFRESH platform uniquely enables stable multimaterial embedded bioprinting of two distinct, tissue-specific dECM hydrogels, thereby extending the applicability of yield-stress bath printing to structurally complex, biologically specialized matrices.

An acellular construct printed using alternating dCM–dMM inks further confirmed the approach, demonstrating high shape fidelity and uniform internal layering without delamination or collapse during or after leaching (Figure 7e) [50]. These findings show that the REFRESH platform enables high-fidelity co-patterning of multiple, tissue-specific dECM bioinks within a single, self-supporting construct, thereby advancing embedded bioprinting toward the fabrication of architecturally and biologically zoned soft-tissue models.

We next applied the REFRESH platform to fabricate a biomimetic multilayer tracheal construct that reproduces the native spatial organization of airway memetic tissue. Using embedded bioprinting, chondrocyte-laden dCM was patterned to form the cartilage (C) region, while NIH-3T3-laden dMM defined the

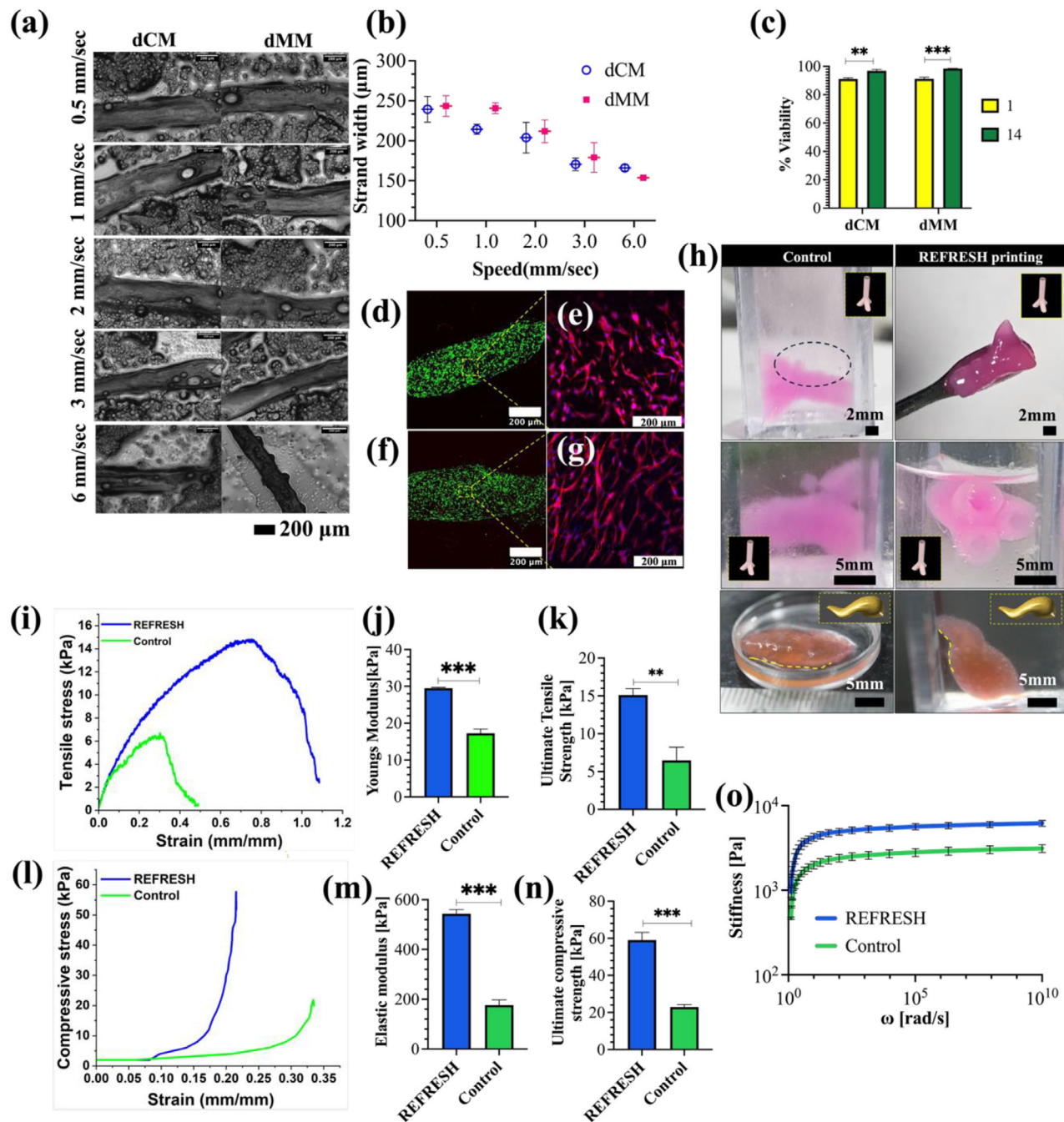


FIGURE 6 | Structural fidelity, biocompatibility, and mechanical performance of constructs fabricated by REFRESH embedded bioprinting. (a) Microscopic images of stand analysis at different print speeds during embedded printing. Scale bar = 200 μm (b) Quantification of strand width across dCM and dMM-based constructs during embedded printing, (c) % viability of bioinks during embedded bioprinting with NIH 3T3 cells encapsulated on days 1 and 14. Statistical significance was determined by an unpaired two-tailed Student's t test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant). Live/dead staining (Calcein AM/PI) (scale bar = 200 μm) and cytoskeleton imaging (Phalloidin/Hoechst) (scale bar = 200 μm) of bioprinted samples demonstrating the biocompatibility and cellular alignment after printing on Day 14 for: (d, e) dCM, (f, g) dMM. (h) Embedded bioprinting of a bronchial branch model tongue-shaped construct in REFRESH bath (5P-10G) and control (0P-10G) using dECM hydrogel ink. Insets show corresponding CAD models. (Scale bar = 2, 5 mm) (i) Representative tensile stress-strain curves of REFRESH-printed and control constructs, showing enhanced tensile strength and extensibility following embedded printing. (j, k) Quantification of tensile mechanical properties, including tensile modulus and ultimate tensile strength, confirming mechanical reinforcement in REFRESH-printed constructs. Statistical significance was determined by an unpaired two-tailed Student's t test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant). (l) Representative compressive stress-strain curves comparing REFRESH and control samples, demonstrating improved resistance to compressive deformation. (m, n) Quantification of compressive modulus and maximum compressive stress, highlighting superior compressive performance achieved via REFRESH embedded printing. Statistical significance was determined by unpaired two-tailed Student's t test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant). (o) Oscillatory frequency sweep-derived stiffness of REFRESH-printed and control constructs, showing higher storage modulus and enhanced dynamic mechanical stability across the tested frequency range. Data are presented as mean $\pm$ SD ($n = 3$).

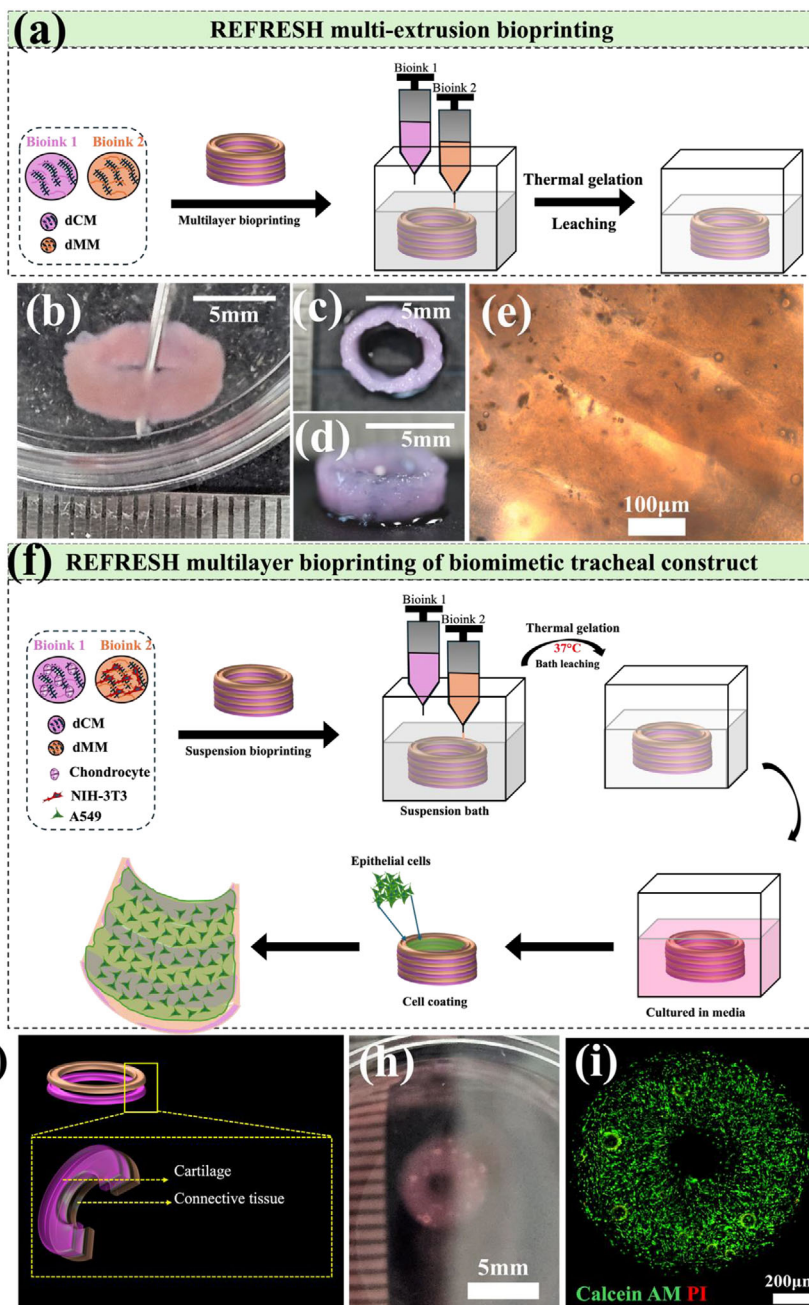


FIGURE 7 | Optimization of embedded bioprinting for multilayered tracheal tissue constructs using a microgel suspension bath and bioprinting of a biomimetic tracheal construct. (a) Schematic workflow of microgel bath-assisted dual-extrusion bioprinting with two distinct hydrogel inks for tracheal submucosa tissue modeling. (b–d) Photographs of printed trachea-mimetic constructs, demonstrating structural stability, hollow lumen formation, and multilayer fidelity post-printing. (Scale bar = 5 mm) (e) Bright-field image showing microscale features and cell-material integration within printed constructs after crosslinking and bath removal. (Scale bar = 100 μm). (f) Schematic illustration of the suspension bioprinting strategy used to fabricate multilayered tracheal tissue constructs. Two bioinks—dCM (decellularized cartilage matrix) and dMM (decellularized muscle matrix)—loaded with chondrocytes, NIH-3T3, and A549 cells were co-printed using a dual-extrusion system into a PEG-Gelatin microgel bath. After thermal gelation (37°C) and suspension bath removal, constructs were cultured and seeded with epithelial cells to model the mucosal layer. (g) Schematic showing region-specific spatial distribution of cartilage and connective tissue mimics within the bioprinted biomimetic trachea. (h) Photograph of the printed ring-shaped construct submerged in media. (Scale bar = 5 mm). (i) Live/dead staining of the construct on Day 10 showing high cell viability (Calcein AM: green; PI: red), (Scale bar = 200 μm).

adjacent connective-tissue (CT) layer (Figure 7f,g). Following thermal gelation, bath removal, epithelial coating, and culture, the constructs retained cylindrical geometry with sharply segregated C–CT interfaces, demonstrating that REFRESH enables spatially resolved deposition of compositionally distinct dECM bioinks without interfacial dilution (Figure 7h). Sustained high cell viability and progressively increasing metabolic activity over 14 days (Figure 7i, Figure S14) further confirmed the cytocompatibility of both the printing microenvironment and the dECM matrices.

2.10 | REFRESH-Enabled Embedded Bioprinting of a Biomimetic Upper Respiratory Tract Tracheal Construct

The multilayered tracheal construct was designed to mimic the native architecture with distinct cartilage (C) and connective tissue (CT) regions (Figure 8a); a schematic summary of the hydrogel–cell combinations (dCM with chondrocytes for cartilage and dMM with NIH 3T3 fibroblasts for connective tissue) is presented in Figure 8b. Immunofluorescence analysis at Day 14 revealed clear region-specific maintenance of lineage phenotypes (Figure 8c(i–iv)). The CT domain exhibited pronounced vimentin (VIM) expression and elongated fibroblast morphology, whereas the cartilage region showed strong aggrecan (ACAN) and collagen II (COL II) deposition, consistent with chondrogenic stabilization [48]. At the luminal surface, ZO-1 and MUC5AC expression indicated epithelial junctional maturation and mucin-associated differentiation [49]. Composite signal mapping revealed sharply demarcated biological boundaries with minimal cross-regional mixing, indicating preservation of biological compartmentalization alongside geometric fidelity. Quantitative image analysis confirmed significant increases in VIM, COL II, ACAN, and ZO-1 over time (Figure 8d–f), reflecting progressive, region-dependent ECM remodeling and epithelial maturation [49–51].

RT-qPCR profiling further supported these findings, with COL2A1 and ACAN upregulated ~10–14 fold and ECAD and ZO-1 increased ~7–13 fold ($p < 0.001$), confirming reinforcement of cartilage-like matrix synthesis and epithelial barrier functions (Figure 8g) [52]. Moderate MUC5AC elevation suggested mucin-associated phenotype development, while VIM transcription remained largely unchanged, indicating dual epithelial–chondrogenic stabilization without mesenchymal-transition activation [53, 54]. These results are consistent with prior evidence that tissue-specific dECM promotes lineage maintenance and ECM deposition, particularly in cartilage and airway systems [4]. However, earlier work has typically demonstrated these effects in single-tissue constructs or manually assembled layered systems [55, 56]. In contrast, the present study demonstrates simultaneous embedded patterning of two distinct tissue-specific dECM hydrogels into a single, fully integrated, zonally organized airway construct — representing a key advance in multi-dECM tissue fabrication.

While embedded printing has previously been shown to stabilize soft hydrogel architectures, the present findings extend this concept by showing that REFRESH preserves phenotype-specific zonation across adjacent biological compartments formed from

different dECM sources. Overall, REFRESH-assisted multimate-rial bioprinting enables the fabrication of a structurally and biologically compartmentalized tracheal model in which cartilage, stromal, and epithelial layers remain distinct yet functionally integrated. The dual advantage of tissue-specific biochemical signaling and precise spatial definition establishes REFRESH as a next-generation platform for printing hierarchically organized airway constructs.

2.11 | REFRESH Supported Bioprinting of a Biomimetic Lower Respiratory Tract: Trifurcate Bronchial Tree Structure

To further evaluate the versatility of the REFRESH embedded bioprinting platform across ECM matrices with distinct rheological and compositional characteristics, a lung-derived dECM (dLuM) hydrogel ink was investigated. Rheological modeling confirmed that dLuM exhibits pronounced shear-thinning behavior (Power-law $n = 0.1668$) with moderate flow resistance ($K = 94.25 \text{ Pa} \cdot \text{s}^n$), indicating strong non-Newtonian characteristics typical of compliant ECM hydrogels (Figure S15a,c). Herschel–Bulkley analysis further revealed a finite yield stress ($\tau_0 = 30.85 \text{ Pa}$) (Figure S15b,d), indicating measurable solid-like behavior prior to flow initiation. These parameters suggest that dLuM possesses sufficient structural integrity to maintain filament definition after deposition while retaining effective shear-thinning for smooth extrusion within the REFRESH bath. Based on these favorable rheological properties, UMSC-laden dLuM bioink was used to fabricate a free-form trifurcated bronchial construct, representing a geometrically demanding lower respiratory tract architecture and serving as a stringent test of bath-mediated confinement and structural stabilization during embedded bioprinting.

Embedded deposition within the REFRESH bath enabled precise definition and geometric retention of all branch segments throughout printing and gelation, without collapse, inter-branch fusion, or distortion—failure modes commonly reported for conventional surface-printed ECM hydrogels (Figure 9a–i). Importantly, constructs were retrieved cleanly following thermal leaching of the bath, confirming that mechanical confinement during printing did not compromise post-fabrication handling. These observations extend previous demonstrations of granular-bath printing for soft hydrogels by showing that branching airway-like topologies can be preserved when the support medium exhibits sufficient yield stress and rapid self-recovery [57].

The printed lung constructs were cytocompatible, along with sustained metabolic activity confirmed by MTT analysis over 14 days (Figure S16). Immunofluorescence imaging revealed dense cell distribution and pronounced α -smooth muscle actin (α -SMA) organization with coordinated nuclear alignment (Figure 9j–l), indicating active cytoskeletal remodeling and emergence of a contractile stromal-like phenotype. CTCF-based image quantification verified elevated α -SMA levels in REFRESH-printed samples (Figure 9m), while RT-qPCR showed a ~4–5 fold upregulation by Day 14 ($p < 0.001$) (Figure 9n). As α -SMA is a hallmark of myofibroblast and smooth muscle cell activation, suggesting mechanosensitive engagement of cytoskeletal signaling driven by the dLuM microenvironment and matrix–cell interactions [58, 59]. Controlled α -SMA expression has

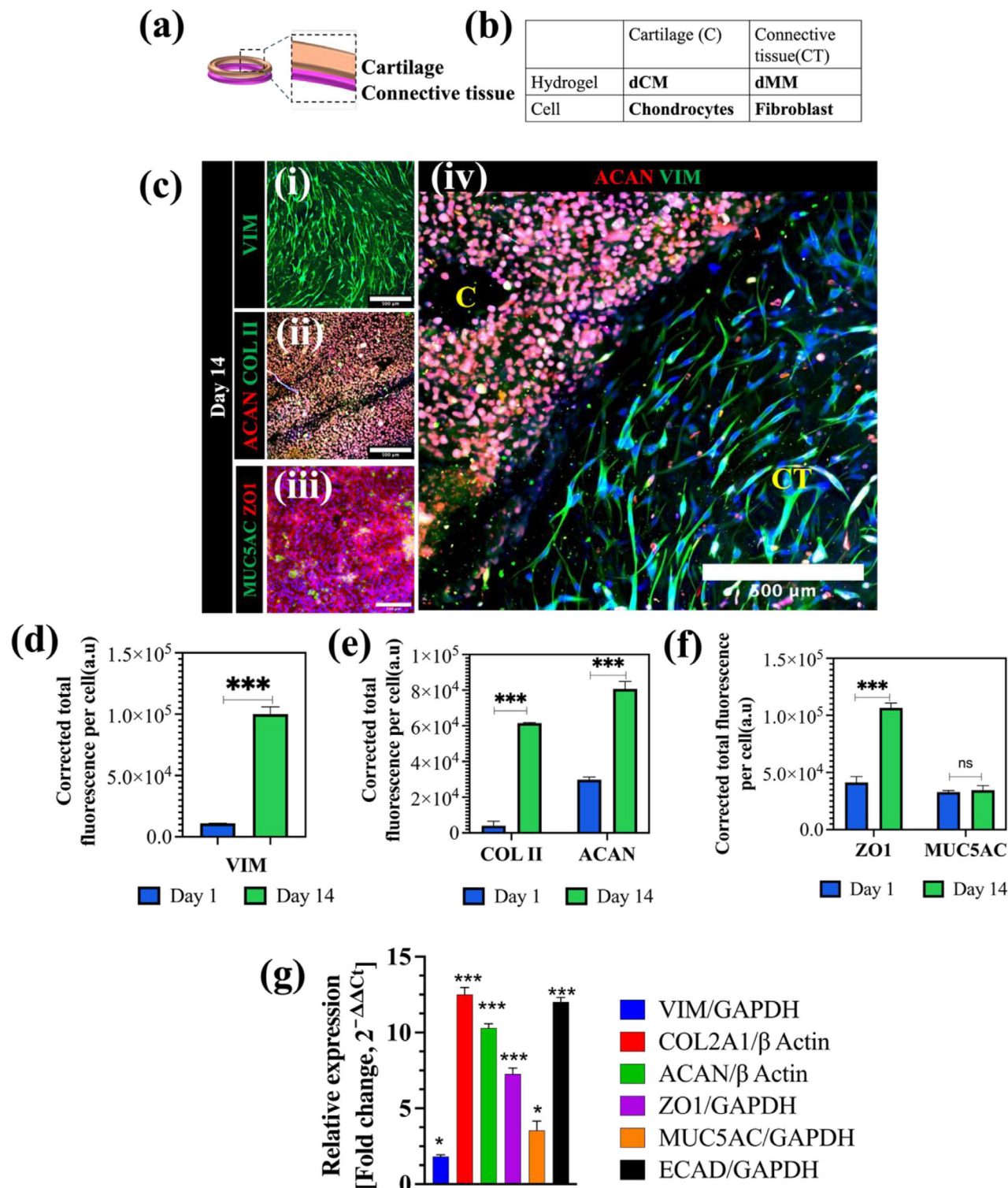


FIGURE 8 | Region-specific maturation in REFRESH-printed multilayer tracheal constructs. (a-b) Schematic showing the spatial organization of the cartilage (C) and connective-tissue (CT) compartments within the multilayer construct and composition. (c) Immunofluorescence staining at Day 14 demonstrating compartment-appropriate phenotypes: (i) Vimentin (VIM) expression in the stromal CT region; (ii) aggrecan (ACAN) and collagen II (COL II) deposition in the cartilage region, (iii) ZO-1 and MUC5AC expression at the luminal epithelial interface, (iv) composite image illustrating sharply defined boundaries between C and CT layers. Scale bars = 500 μm. Quantitative analysis of corrected total cell fluorescence (CTCF): (d) VIM expression in the stromal CT region; (e) aggrecan (ACAN) and collagen II (COL II) deposition in the cartilage region, (f) ZO-1 and MUC5AC. Statistical significance was determined by an unpaired two-tailed Student's t test ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant). (g) RT-qPCR fold-change in gene expression of mesenchymal (VIM), chondrogenic (COL2A1, ACAN), epithelial (ZO-1, MUC5AC, E-cadherin) markers was quantified by RT-qPCR and normalized to GAPDH/ β -actin using the $2^{-\Delta\Delta C_t}$ method. Fold change represents gene expression at Day 14 relative to Day 1 (Day 1 = 1.0). Statistical significance was calculated relative to Day 1 by one-way ANOVA ($n = 3$, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant).

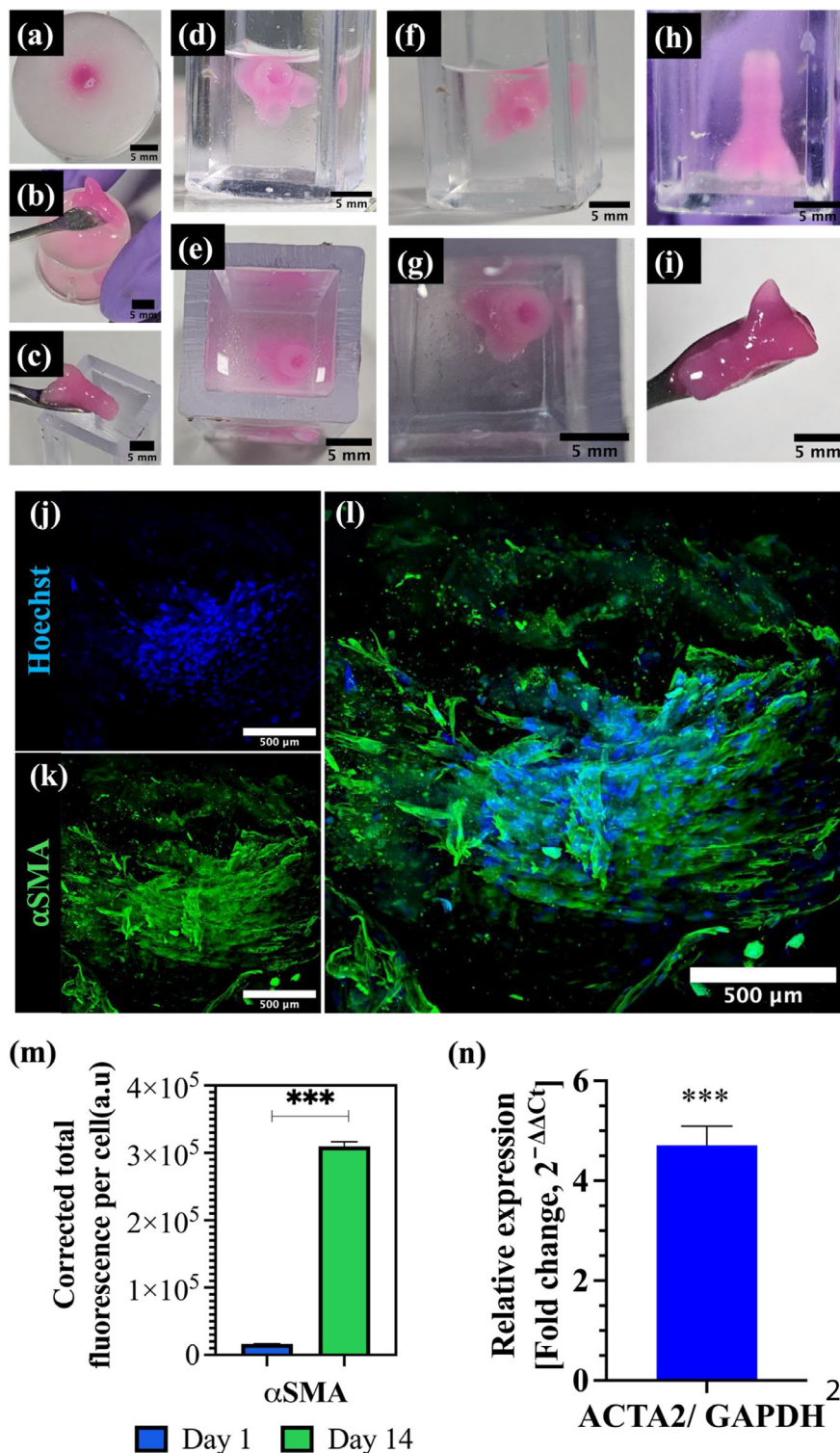


FIGURE 9 | REFRESH-enabled embedded bioprinting and cellular remodeling within a trifurcated bronchial construct. (a–c) Photographs showing the embedded printing of a trifurcated bronchial construct using lung dECM (dLuM) hydrogel ink within the PEG–gelatin REFRESH suspension bath. (d–g) The printed construct retained its free-form geometry during extrusion and post-printing incubation, demonstrating stable maintenance of complex branch topology without collapse, spreading, or inter-branch fusion. (h,i) Following gentle thermal removal of the suspension bath, the construct was retrieved intact with high geometric fidelity and structural continuity. Immunofluorescence imaging demonstrated (j) Hoechst staining revealed widespread nuclear localization, (k) αSMA staining confirmed cytoskeletal activation, and the merged image (l) showed spatially coordinated alignment of αSMA-positive cells within the dLuM matrix. Scale bars = 500 μm. (m) Quantitative image analysis of corrected total cell fluorescence (CTCF) demonstrated significantly elevated αSMA expression in REFRESH-printed constructs, (n) RT-qPCR fold-change in gene expression of αSMA quantified by RT-qPCR and normalized to GAPDH using the 2^{-ΔΔCt} method. Fold change represents gene expression at Day 14 relative to Day 1 (Day 1 = 1.0). Statistical significance was calculated relative to Day 1 by unpaired two-tailed Student's t test (n = 3, $p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant).

been associated with constructive stromal remodeling and tissue stabilization, whereas persistent overactivation may predispose toward matrix stiffening or fibrosis [58]. Thus, while the observed stromal activation may contribute beneficial mechanical reinforcement, future studies will be needed to tune biochemical and mechanical cues to balance constructive remodeling with fibrosis avoidance. Taken together, these results demonstrate that REFRESH-assisted embedded bioprinting enables the fabrication of airway constructs that combine high structural fidelity with the emergence of mechanically active stromal phenotypes within lung-derived ECM — capabilities that are difficult to achieve using conventional surface-extrusion approaches for soft ECM hydrogels. Accordingly, REFRESH provides a robust route for generating anatomically organized and biologically responsive airway models with relevance for regenerative engineering.

2.12 | Tissue-Specific Versatility of REFRESH Embedded Bioprinting With Diverse dECM Hydrogel Inks

REFRESH embedded bioprinting represents a distinct advance in suspension-bath-assisted fabrication by enabling structurally stable, free-form printing of minimally modified dECM hydrogels from multiple organ sources. Using REFRESH, decellularized matrices derived from cornea (dCoM), kidney (dKM), submucosa (dSM), muscle (dMM), skin (dSkM), cartilage (dCM), liver (dLM), and lung (dLuM) were printed into complex 3D geometries that reproduced essential features of native anatomy (Figure 10a–h). To mechanistically assess the compatibility of these diverse matrices with embedded printing, viscosity and shear-stress profiles were fitted using Power-law and Herschel–Bulkley models (Figure S17a–d). All formulations exhibited pronounced shear-thinning behavior ($n = 0.166–0.319$), confirming non-Newtonian flow across tissues. The consistency index (K) varied substantially between matrices, reflecting intrinsic differences in ECM composition and density, with dKM displaying the highest low-shear resistance ($K = 231 \text{ Pa} \cdot \text{s}^n$) and dLM the lowest ($K = 10.26 \text{ Pa} \cdot \text{s}^n$). Herschel–Bulkley analysis revealed finite yield stresses ($\tau_0 = 2.95–27.82 \text{ Pa}$), indicating that each matrix possessed measurable solid-like structure prior to flow initiation. Notably, despite this wide rheological range, all inks maintained sufficient shear-thinning for smooth extrusion while preserving post-deposition filament integrity within the REFRESH bath.

Crucially, all constructs were formed from low-viscosity, non-reinforced dECM hydrogels, demonstrating that REFRESH uniquely decouples geometric fidelity from intrinsic ink rheology. Here, we try to address a central limitation of conventional extrusion printing, where soft ECMs typically spread, collapse, or fuse during deposition. Across the tissue spectrum, REFRESH supported transparent corneal domes, lobular kidney-like forms, submucosal cylinders, layered muscle constructs, tongue from skin, tooth mimicking structure from cartilage, and trifurcate structure. All constructs remained free-standing following thermoreversible bath removal, confirming robust stabilization without chemical cross-linking or rheology-modifying additives. This breadth of compatibility highlights the core novelty of the REFRESH bath: a PEG–gelatin microgel support medium engineered to combine yield-stress-dominated confinement with rapid thixotropic recovery and complete thermal reversibility.

These properties ensure localized flow during extrusion and immediate re-solidification around the filament, preventing spreading, sagging, or interlayer fusion—failure modes consistently reported for surface and weak-support printing. While previous granular baths improved printing of soft matrices, most have been restricted to synthetic or single-source ECM systems and have not demonstrated robust performance across diverse native dECM formulations [60–62]. REFRESH functions as a broadly tissue-agnostic platform, supporting ECM hydrogels spanning cornea, kidney, submucosa, muscle, skin, cartilage, liver, and lung origins without formulation-specific optimization. Furthermore, the thermoreversible and biocompatible REFRESH matrix allows clean, damage-free construct retrieval, avoiding the abrasive handling or chemical exposure associated with Carbopol- or agarose-based baths, thereby preserving the ultra-structure and bioactivity of fragile ECM networks.

Together, these findings position REFRESH as a new class of suspension bath — one that transforms embedded bioprinting from a material-limited technique into a generalizable fabrication framework for native, tissue-derived ECM hydrogels. By enabling high-resolution, free-form printing of biologically authentic dECM across multiple organ types, REFRESH substantially expands the design space for organ-specific regenerative constructs, physiologically relevant disease models, and personalized tissue manufacturing.

3 | Experimental Section

3.1 | Preparation of PEG–Gelatin Microgel Slurry Suspension Bath and Its Characterization

The PEG–Gelatin microgel slurry suspension bath was prepared via a water-in-oil (W/O) emulsification method. Briefly, Gelatin Type B (Bloom strength 150, SRL) was initially dissolved in warm distilled water at 40–50°C to prepare solutions with final concentrations of 5% and 10% (w/v). Simultaneously, polyethylene glycol (PEG, MW 20,000, SRL, India) was solubilized in warm distilled water at 50°C to prepare a final concentration of 5% (w/v) solution. The PEG solution was subsequently combined with the gelatin solution in a 1:1 ratio while maintaining continuous agitation until a homogenous aqueous phase was achieved and abbreviated as 5P-5G, 5P-10G. The emulsion was created by combining the aqueous phase with Paraffin Liquid Heavy (Mineral oil heavy, Extra pure, 110–230 mPas, SRL, India), consisting of 0.1% Tween 20 (HiMedia, India) as a surfactant. The oil phase was mixed into the aqueous phase at a 2:1 (v/v) ratio with continuous agitation, and the mixture was emulsified with a high-speed homogenizer to produce fine aqueous droplets. The resultant emulsion was subsequently cooled to promote the gelation of the gelatin at 4°C for 10 min, resulting in the formation of microgels. The PEG–Gelatin microgels were subsequently isolated from the oil phase using centrifugation at 1000 rpm for 5 min at 15–18°C. The collected microgels were extensively rinsed with cold distilled water multiple times by centrifuging at 9000 rpm, to eliminate residual oil and surfactant, resulting in the final transparent microgel slurry suspension bath. For cell culture-based studies, a sterile suspension bath was used. Sterile suspension bath preparation is mentioned in the Section S11.

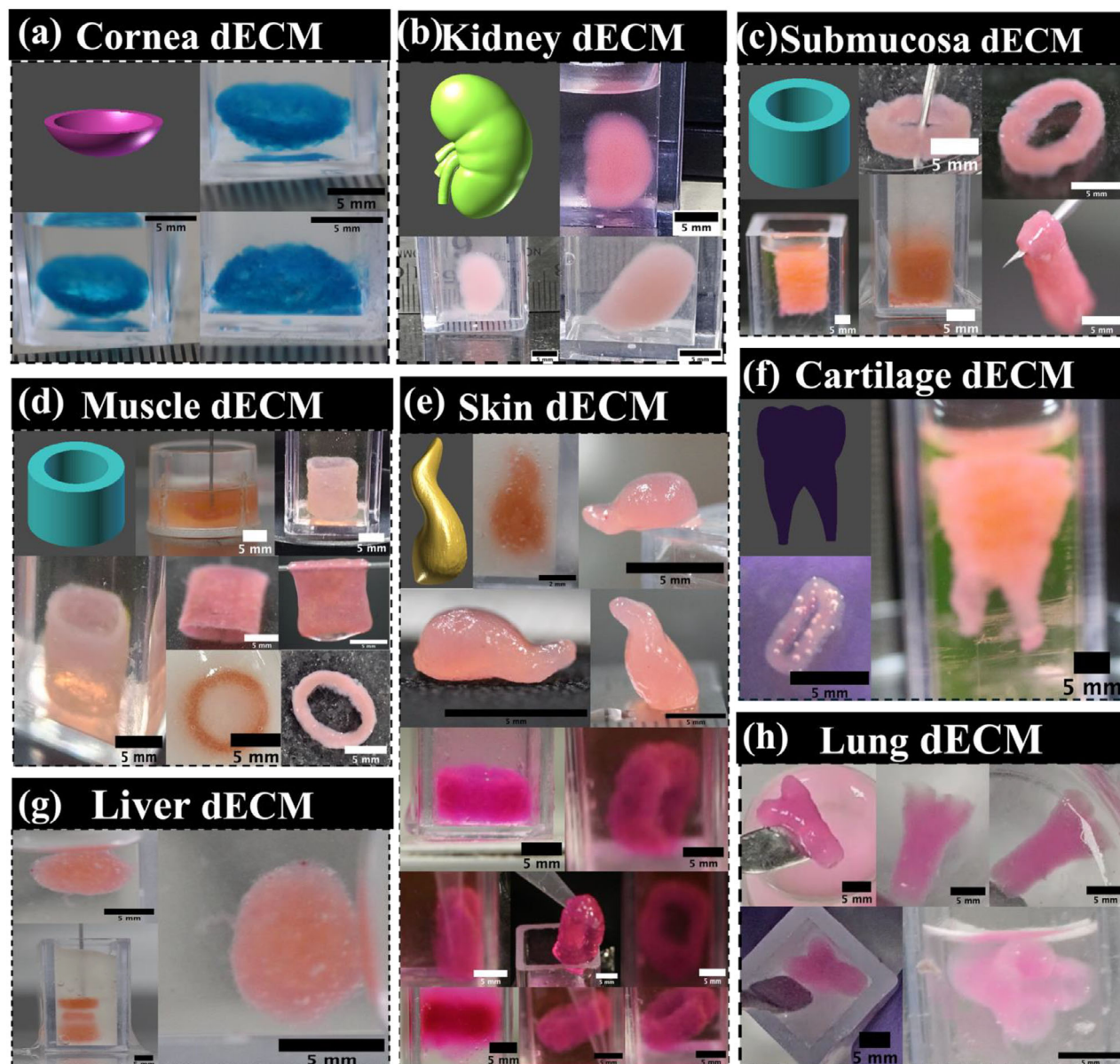


FIGURE 10 | Representative images of tissue-specific decellularized extracellular matrix (dECM) hydrogel ink constructs fabricated using embedded bioprinting within a PEG-Gelatin microgel suspension bath. (a) Cornea-derived dECM showing transparent, dome-shaped constructs mimicking native corneal curvature, (b) Kidney-derived dECM bioprinted into solid and lobular forms, reflecting renal-like geometry, (c) Cylindrical structure printed using submucosa-derived dECM mimicking tubular organs, (d) Muscle-derived dECM exhibiting cylindrical and layered constructs with high print fidelity, and structural stability, (e) Skin-derived dECM printed into tongue- and cylindrical-like constructs, highlighting adaptability to complex surface geometries; (f) Cartilage-derived dECM fabricated into anatomically inspired form—a tooth—demonstrating robust shape retention; (g) Liver-derived dECM constructs showing solid and multi-layered structures with homogeneous distribution, (h) Lung-derived dECM printed into branching and lobe-like structures, capturing characteristic organ-specific architecture. Scale bars = 5 mm.

3.1.1 | Gelatin-Only Control Suspension Bath

Gelatin-only microgel suspension baths (OP:10G) were prepared using the same water-in-oil (W/O) emulsification protocol described above, using 10% (w/v) gelatin as the aqueous phase, with PEG omitted. All processing parameters—including oil-to-aqueous phase ratio, homogenization conditions, cooling-induced gelation, and washing steps—were maintained identical to those used for the PEG-gelatin microgel suspension bath to ensure direct comparability.

3.1.2 | Rheological Analysis of Microgel Slurry Suspension Bath

The rheological properties of the PEG-Gelatin microgel slurry suspension bath were investigated using the Malvern Kinexus pro rheometer (United Kingdom), equipped with a 20 mm parallel plate geometry with a gap of 0.3 mm at 25°C. To analyze the viscosity profile and yield stress of the microgel, an equilibrium flow curve was performed with shear rates from 0.1% to 1000%. Self-recovery behavior was analyzed using the thixotropy test

with lower and higher strain %. A time-dependent temperature sweep test was done from 4°C to 37°C to analyze the thermal sensitivity of the microgels. Once the samples reached 37°C, an isothermal time sweep was performed for 45 min. A frequency sweep was performed to analyze the jammed behavior. Amplitude sweep was analyzed to determine the linear viscoelastic region (LVER) of the microgel. Analysis was performed for all test samples, 5P-5G, 5P-10G, in comparison to the control 0P-10G in triplicate.

3.1.3 | Particle Size and Size Distribution

The microgel obtained was characterized using an optical microscope (Leica DMI1) with a 20X objective for analysis of the gel particle size distribution. For SEM analysis, PEG-Gelatin microgel slurry and control slurry were subjected to freeze-drying and subsequently gold sputtered using a thin film deposition machine (Kurt J. Lesker CMS-18 HV) for 60 s with 25 mA electric current, and examined using scanning electron microscopy (SEM; JOEL JSM-7800F, Japan). Particles were analyzed using the ImageJ software, and a histogram of particle size was analyzed.

3.1.4 | Characterization of Thermal Responsiveness and Optical Transparency of Microgel Slurries

To evaluate the thermal reversibility of the microgel-based suspension baths, 5P-5G and 5P-10G baths were prepared and incubated at two temperatures: room temperature (25 °C) and physiological temperature (37 °C). Each slurry was loaded into a 12-well plate, visually inspected, and images were captured and compared with the gelatin-only microgel control(0P-10G).

3.1.5 | Optical Transparency

The optical transparency of 5P-5G and 5P-10G bath and control 0P-10G was quantified using ultraviolet-visible (UV-vis) spectroscopy (JASCO V-730). Slurries were loaded into 1 cm pathlength quartz cuvettes and scanned across a wavelength range of 200–800 nm using a microplate spectrophotometer (JASCO V-730). Absorbance values were recorded and plotted to assess light transmittance characteristics. All measurements were performed using $n = 3$ independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.2 | Hydrogel Inks Preparation and Characterization

3.2.1 | Isolation of Caprine dCM, dMM

Caprine trachea tissue was obtained from a nearby local source. The tissue was thoroughly washed and cleaned of fat deposits. The tissue was further separated into cartilage and trachealis muscle present dorsally and treated for decellularization separately. Both tissues were chopped into small pieces of about 2–3 mm and subjected to a detergent-free salt-based decellularization process. Both tissues underwent the detergent-free decellularization process separately, as mentioned elsewhere [36]. Briefly,

- i. Tissue chopped into very small pieces, about 2–3 mm in size
- ii. Treated with hypertonic, hypotonic solution of NaCl for 4 h alternatively, followed by a distilled DI water wash for about 30 min between the alternative hypertonic and hypotonic solution change.
- iii. Continued the process for 3 days, followed by treatment With 3% H₂O₂ for 30 min and rinsed with DI water
- iv. Tissue washed with distilled water for 2 days by changing the water every 5 h.
- v. The decellularized tissue is drained of excess water and kept at -20°C overnight and further subjected to lyophilization.
- vi. Lyophilized sample stored at -20°C until further use.

3.2.2 | Histological Analysis

Post-decellularization, both native and decellularized tissues were sectioned using a microtome at 7 μ m thickness. The sections were stained with Hematoxylin and Eosin (H&E) to evaluate overall tissue architecture and with Hoechst 33258 (DAPI) to assess the efficiency of decellularization by examining nuclear staining.

3.2.3 | Biochemical Characterization

Native and decellularized tissues were biochemically characterized to evaluate the efficiency of decellularization by quantifying DNA, glycosaminoglycan (GAG), and collagen contents. For all biochemical assays, tissues were digested in 1 mL of papain solution for 16 h at 60°C, following a previously reported protocol [36]. Briefly, the papain solution consisted of 125 μ g mL⁻¹ papain prepared in 0.1 M sodium phosphate buffer containing 5 mM EDTA and 5 mM cysteine-HCl (pH 6.5). All measurements were performed using three independent biological replicates ($n = 3$), with each biological replicate analyzed in technical triplicates to ensure reproducibility.

DNA content was quantified using the PicoGreen dsDNA assay kit (Invitrogen, UK). Fluorescence intensity was measured using an EnSpire 2300 multimode spectrophotometer, and DNA concentration was calculated from a standard curve generated using calf thymus DNA. sGAG content was determined using the 1,9-dimethylmethylene blue (DMMB) assay. Absorbance was measured at 492 nm using an EnSpire 2300 multimode plate reader, and sGAG concentration was quantified against a chondroitin sulfate standard curve. Collagen content retained in the decellularized tissues was assessed via a hydroxyproline assay using chloramine-T and dimethylaminobenzaldehyde (Sigma-Aldrich, USA) according to the manufacturer's protocol. Absorbance was measured at 550 nm using an EnSpire 2300 multimode plate reader, and collagen content was calculated from a hydroxyproline standard curve.

3.2.4 | Hydrogel Preparation

Pregels were prepared with dCM and dMM by weighing the required amount of lyophilized tissue, ball-milled (5 min, 30 Hz

frequency), and underwent pepsin digestion by solubilizing in a solution containing 0.1 M HCl with 10% (w/v) pepsin with respect to the weight of the tissue. After solubilization for 18 h, the pregel was pH adjusted to 7.4 by adding 10 M ice-cold NaOH. To prevent gelation throughout the pH adjustment procedure, the pregel was always maintained at a temperature below 4°C. This was kept at 4°C in a refrigerator until further use. The hydrogel was formed by incubating the neutralized pregel at 37°C.

3.2.5 | Swelling Study of Hydrogels

To measure the water diffusion into the hydrogel over time, a swelling study was conducted. To achieve a consistent amount of gels, all pH-adjusted hydrogels were weighed and allowed to gel for 30 min at 37°C in a cylinder-shaped mold. Following gelation, it was submerged in PBS and stored in an incubator with 5% CO₂ at 37°C. At different intervals, the hydrogels were taken out of PBS and weighed after the excess PBS was blotted using moist tissue paper. The following formula was used to calculate the results:

$$Q = \frac{(M_s - M_i)}{M_i}$$

Here, Q is the swelling index, M_s is the mass in the swollen state, and M_i is the mass in the initial state. All measurements were performed using n = 3 independent samples per group, and data were reported as mean ± standard deviation.

3.2.6 | Rheology of Hydrogel

Rheological tests for the pH-adjusted pregel were done on a rheometer (Kinexus Pro+, Malvern Panalyticals) equipped with a 20 mm parallel plate geometry. A fixed gap of 0.3 mm was employed to ensure sufficient torque sensitivity for soft dECM hydrogels while minimizing wall slip, sample dehydration, and edge effects during testing. Sample dehydration was prevented by enclosing the system in a solvent trap and applying mineral oil to the fixtures' edges. Shear sweep from 0.1 to 1000/s was performed on the dECM pregel at a temperature of 15°C to analyze the pregel viscosity. Shear recovery profile of the hydrogels was evaluated at a temperature of 15°C with high and low shear strain for 30 min. We performed a 1 h time sweep at a frequency of 1 Hz and a strain of 0.1% to comprehend the moduli of pregel at 37°C for an extended period. A temperature sweep was performed from a temperature range of 4°C to 37°C to understand the dynamic moduli of the hydrogel at a crosslinking temperature of 37°C. Once the samples reached 37°C, an isothermal time sweep was performed for 45 min. Strain-dependent storage and loss moduli of the pregel were measured by performing an amplitude sweep in the range of 0.1%–1000% at 15°C. Additionally, frequency sweep analysis was used to determine frequency-dependent storage and loss moduli at 1% strain in the 0.1–100 Hz range. All measurements were performed using n = 3 independent samples per group, and data were reported as mean ± standard deviation.

3.3 | Turbidity Gelation Kinetics Assay

Gelation kinetics of the dECM formulations were evaluated by turbidimetry as previously described in the literature, with

minor modifications [39]. Neutralized pre-gel solutions (final concentration: 0.1% w/v dECM) were prepared on ice and divided into four experimental groups: (i) dECM only, (ii) dECM + 5% (w/v) PEG, (iii) dECM + 2 mM urea + 5% (w/v) PEG, and (iv) dECM + 2 mM urea. A total of 100 µL of each formulation was dispensed in triplicate into UV-transparent 96-well plates, and PBS-filled wells were included for background subtraction. The optical density at 405 nm was recorded using a temperature-controlled microplate reader during incubation at 37°C. Measurements were acquired every 1–2 min for 60–90 min, until a clear plateau in turbidity was observed. Absorbance data were blank-subtracted and normalized to the initial baseline, and gelation parameters including lag time, gelation half-time (t_{1/2}), and maximum turbidity (A_{max}) were extracted from the resulting sigmoidal curves. All measurements were performed in triplicate.

3.4 | Oscillatory Rheology Test

Time-temperature-dependent gelation of the dECM hydrogels was characterized using oscillatory rheology (Malvern Kinexus pro rheometer, United Kingdom), equipped with a 20 mm parallel plate geometry. Acid-solubilized dECM pre-gel solutions were loaded onto a temperature-controlled parallel-plate geometry (gap = 0.3 mm), and the pH was adjusted to the physiological range directly on the rheometer plate. For the PEG-containing samples, 5% (w/v) PEG was added immediately after pH adjustment and mixed to obtain a homogeneous pre-gel. A small-amplitude oscillatory strain within the linear viscoelastic region and a fixed frequency (0.5 Hz) were applied throughout the test. The temperature was first ramped from 4°C to 37°C, during which the complex viscosity (η*) was recorded continuously to capture the sol-gel transition. Once the samples reached 37°C, an isothermal time sweep was performed for 45 min to monitor continued network maturation under physiological conditions. All measurements were conducted in triplicate for both dECM-only and dECM + 5% PEG groups. All measurements were performed using n = 3 independent samples per group, and data were reported as mean ± standard deviation.

3.5 | Microstructure of the Printed Hydrogels

The microstructure of the printed hydrogels was examined by scanning electron microscopy. Constructs printed within the REFRESH suspension bath and the control suspension bath (OP-10G) were first incubated at 37°C to enable bath liquefaction and complete leaching of the supporting matrix. The hydrogels were then gently rinsed with PBS to remove residual bath components. Samples were dehydrated through a graded ethanol series (30%, 50%, 70%, 90%, and 100%; 15 min at each step), frozen, and lyophilized. Dried constructs were mounted on metal stubs and sputter-coated with gold for 60 sec (Kurt J. Lesker CMS-18 HV, 60 sec) and examined using scanning electron microscopy (SEM; JOEL JSM-7800F, Japan).

3.6 | Printing Optimization of Hydrogels in the Microgel Suspension Bath

Embedded 3D printing was performed using a commercial 3D printer (Trivima, NBIL, Bengaluru, India) housed inside a

biosafety cabinet. Printing paths were designed in CAD software, exported as .stl files, and converted to G-code using Simplify3D. The final G-code was executed using Pronterface control software. The pregel was loaded into a 3-mL printing cartridge fitted with a 330- μ m nozzle, and all printing was conducted at 15°C. For printing studies, 6% (w/v) dCM hydrogels were prepared, centrifuged at 4000 rpm for 10 min at 4°C to remove entrapped air, and the pH was adjusted to the physiological range prior to use. Straight filaments were extruded within the microgel suspension bath ((5P-5G, 5P-10G) with respect to control (0P-10G) while maintaining a constant extrusion pressure. To evaluate the effect of deposition rate on print fidelity, the printing speed was varied systematically at 0.5, 1, 3, 5, 9, 12, and 15 mm s⁻¹, with a minimum of three strands printed per condition. Following printing, the constructs were leached at 37°C imaged in situ using an optical microscope, and strand fidelity and geometry were qualitatively assessed. Quantitative strand-width measurements were subsequently obtained from the images using ImageJ. All measurements were performed using n = 3 independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.7 | 3D bioprinting Optimization and Characterization

3.7.1 | Printing Optimization of dCM, dMM Hydrogels

A commercial 3D printer, NBIL (Bengaluru, India) Trivima, equipped in a biosafety cabinet (BSC), was used for the bioprinting process. The .stl file was made and converted to G-code by Simplify 3D software, and the file was loaded into the printing software Pronterface. A 3cc cartridge with a 330 μ m nozzle diameter was used throughout the printing process at 15°C.

6% dCM and 5% dMM hydrogels used for printing were filtered (4000 rpm, 10 min, 4°C), pH adjusted, and used for printing. A lattice model was printed with 2 layers (15 $\times$ 15 $\times$ 1) mm. Extrusion-based 3D printing was done layer-by-layer at different flow speeds. After printing, constructs were examined under an optical microscope to determine their strand fidelity.

3.7.2 | Cell Culture

NIH-3T3 and A549 cell lines were procured from NCCS, Pune. Umbilical cord-derived stem cells were procured from Sigma-Aldrich, USA. A549 was cultured in DMEM/F12 (HiMedia, India), NIH-3T3 was cultured in DMEM-high glucose (HiMedia, India), UMSCs were cultured in Knock-out DMEM (Gibco, USA) supplemented with 10% fetal bovine serum (Gibco, USA) in a 37°C, 5% CO₂ humidified environment. Confluent cells were removed from the TCP with 0.25% trypsin solution, centrifuged at 1800 rpm for 5 min, suspended in media, and used for further studies at the desired concentration.

3.7.3 | Chondrocyte Isolation

Primary chondrocytes were isolated from porcine articular cartilage using an optimized enzymatic digestion protocol. Briefly, cartilage tissue was aseptically harvested from the femoral

condyles of fresh porcine knee joints and immediately minced into fine pieces (~1–2 mm³). The tissue fragments were washed three times with sterile PBS supplemented with 1% penicillin-streptomycin. After proper washing, a filtered collagenase solution was prepared with 0.2% collagenase A dissolved in 50% PBS and 50% DMEM without FBS. Solution was further added to the tissue pieces. Incubated for 16 hrs. at 37°C with a humidified atmosphere of 5% CO₂. After the incubation period, the solution was centrifuged at 2000 rpm for 10 min, and the supernatant was discarded. The cell pellets were resuspended in culture medium and allowed to proliferate.

3.7.4 | Hydrogel Inks Preparation and 3D Bioprinting

Prior to the printing, all the parts, including the nozzle, syringe barrel, and plunger, were sterilized with 70% v/v ethanol and exposed to UV radiation for 15 min in a laminar hood. Pregel inks were made over ice to avoid crosslinking and seeded with cells at a density of 10 million cells per mL of hydrogel. Prior to printing, the hydrogel ink-loaded syringe barrel was centrifuged at 2000 rpm, 4°C, for 5 min to eliminate air bubbles. Print speed optimization with hydrogel inks was performed with different speeds, with the pre-optimized pressure by bioprinting a lattice structure of 2 layers (15 $\times$ 15 $\times$ 1 mm). Printing pressure was set at 0.4 bars for dCM and 0.3 bars for dMM at different print speeds of 180, 360, and 540 mm/min.

3.7.5 | Cell Viability Evaluation

The live/dead experiment was performed using calcein-AM (Invitrogen, UK) and propidium iodide probe (Sigma Aldrich, Germany) to distinguish living cells (stained green) and dead cells (stained red), respectively, depending on the integrity of the cell membrane. On days 2, 7, and 14, the bioprinted structures were subjected to live/dead procedures at varying rates. Percentage cell viability was calculated with the ImageJ software (NIH, USA). All measurements were performed using n = 3 independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.7.6 | Immunofluorescence Assessment

For confocal imaging to evaluate the cellular morphology, 3D bioprinted constructs with varying speeds were fixed in 4% Paraformaldehyde (PFA) on days 7, 14 of culture. They were then stained with F-actin (Invitrogen, UK) and Hoechst 33258 (Invitrogen, UK), and images captured.

3.7.7 | Rheological Model Fitting

Rheological analysis was carried out on both dECM hydrogels and the microgel support fluid formulations to evaluate the viscoelastic behavior and shear-thinning properties that are essential for extrusion-based bioprinting. Flow behavior was assessed using two often utilized non-Newtonian fluid models: the Herschel-Bulkley model and the Power Law. All measurements were performed using n = 3 independent samples per group, and data

were reported as mean $\pm$ standard deviation.

$$\eta = K\dot{\gamma}^{n-1} \quad (1)$$

where, n = Flow index (Shear Thinning, where n is less than 1), K = Consistency index (Pa.s), η = Viscosity (Pa.s), $\dot{\gamma}$ = Shear rate (s⁻¹).

$$\tau = \tau_0 + K\dot{\gamma}^n$$

where, n = Flow index, K = Consistency index (Pa.s), τ = Shear stress (Pa), τ_0 = Yield stress (Pa), $\dot{\gamma}$ = Shear rate (s⁻¹).

Using a rotational rheometer over a range of shear rates (0.1 to 1000 s⁻¹), rheological data were acquired. Using Python, with nonlinear regression executed using the `scipy.optimize.curve_fit` function, was used to fit the raw shear stress against shear rate data to the power law model.

3.8 | REFRESH Bioprinting Optimization and Characterization

3.8.1 | Optimization of Embedded Printing

Suspension bath was loaded into a 24-well plate, maintained at 25°C or lower, and printing optimization with suspension bath was carried out in a commercial 3D printing equipment, NBIL Trivima (Bengaluru, India) bioprinter. Line Printing at different speeds of 0.5 mm/sec, 1 mm/sec, 2 mm/sec, 3 mm/sec, and 6 mm/sec was carried out with dCM and DMM hydrogels. Post printing, the setup was kept at 37°C for suspension bath leaching and crosslinking of the printed construct. Crosslinked strands were washed with PBS and further examined under a microscope to determine their strand fidelity. The optimized speed and pressure were chosen for the biocompatibility study of suspension bath-assisted bioprinting. Live/dead experiment was performed on days 1 and 14 using calcein-AM (Invitrogen, UK) and propidium iodide probe (Sigma Aldrich, Germany) to distinguish living cells (stained green) and dead cells (stained red), respectively. The detailed process of bath leaching and removal is described in the Section S8.

3.8.2 | Biocompatibility Study of Embedded Bioprinting

On days 1 and 14, the bioprinted structures were subjected to live/dead procedures. Percentage cell viability was calculated with the ImageJ software (NIH, USA). To evaluate the cellular morphology of suspension bath-assisted 3D bioprinted constructs, the day 14 samples fixed in 4% PFA were stained with F-actin (Invitrogen, UK) and Hoechst 33258 (Invitrogen, UK), and images were captured. All measurements were performed using $n = 3$ independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.9 | Mechanical Characterization of Printed Constructs

3.9.1 | Uniaxial Tensile Testing

Rectangular-shaped hydrogel specimens (acellular) were printed using the REFRESH suspension bath and the control suspension

bath and subsequently incubated at 37°C to enable bath leaching and complete crosslinking. Samples were trimmed to uniform geometry, and their gauge dimensions were measured immediately prior to testing. Uniaxial tensile testing was performed using a mechanical tester equipped with a 10N load cell under hydrated conditions at room temperature. Specimens were strained at a constant rate, and stress-strain curves were recorded. The Young's modulus was calculated from the linear region of the stress-strain curve, while the ultimate tensile strength and strain at failure were determined from the peak load prior to fracture. All measurements were performed using $n = 3$ independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.9.2 | Compression Testing

Cylindrical hydrogel samples (acellular) prepared as described above were subjected to compression testing using a mechanical testing system (Instron Tensile Force Tester) fitted with a 10 N load cell. Samples were placed between two parallel compression platens and compressed at a constant displacement rate of 0.1 mm min⁻¹ until 40% strain. Engineering stress-strain curves were generated from the recorded load-displacement data after normalization to the initial sample dimensions. The compressive (elastic) modulus was calculated from the linear region of the stress-strain curve (typically 10–20% strain). The ultimate compressive strength was defined as the maximum stress sustained by the hydrogel prior to structural collapse. All measurements were performed using $n = 3$ independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.9.3 | Oscillatory Frequency-Sweep Rheology

Dynamic mechanical behavior was assessed using a rotational rheometer equipped with a parallel-plate geometry (gap = 0.3 mm) using Anton Paar, Austria. REFRESH-printed and control-bath-printed constructs (acellular) were tested after complete bath removal at 37°C. Amplitude oscillatory strain within the linear viscoelastic region was applied, and frequency sweeps were performed from 0.1 to 10 rad s⁻¹. The storage modulus (G') and loss modulus (G'') were recorded as a function of angular frequency. All measurements were performed using $n = 3$ independent samples per group, and data were reported as mean $\pm$ standard deviation.

3.10 | Multimaterial Embedded Bioprinting

Multilayered tubular structures were fabricated with a dual-nozzle extrusion bioprinter. Hydrogel inks 1 (6% dCM-hydrogel) and Hydrogel inks 2 (5% dMM-hydrogel) were assigned different print heads and maintained at 15°C to avoid gelation of the hydrogel. The inks were co-printed in alternating layers within the microgel suspension bath to create a tubular structure. Post-printing, the structures underwent thermal gelation at 37°C to thermally crosslink the structure. The microgel suspension bath was subsequently extracted using warm phosphate-buffered saline (PBS), resulting in the freestanding structure. A macroscopic evaluation of the printed structures was carried out

to assess their integrity and shape. Optical microscopy was employed to evaluate the integration of layers and examine layer adherence within the construct.

3.11 | Embedded 3D Bioprinting of Upper Respiratory Multilayered Biomimetic Tracheal Structure

A 3D biomimetic tracheal construct was bioprinted in the suspension bath with multilayers of dCM and dMM printed in alternate layers with dual cartilage. dCM encapsulated with chondrocytes serves as the bottom layer, followed by dMM encapsulated with NIH 3T3. A CAD model for printing the same was prepared using Fusion 360, and the .stl file was converted to G-code by Simplify 3D software. Further, the printing G-code file was loaded into the printing software Pronterface. A 3cc cartridge with a 330 μ m nozzle diameter was used throughout the printing process at 15°C. A scaled-down version of the tracheal model was designed with dimensions of 3 mm in height. After maturation of the printed construct, A549 cells were coated on the lumen side and cultured for 14 days. The suspension bath used for all cell culture-based studies was prepared under sterile conditions and pH adjusted. Cell viability was evaluated with live/dead analysis using calcein-AM (Invitrogen, UK) and a propidium iodide probe (Sigma Aldrich, Germany). Additionally, functional markers were evaluated with Aggrecan, Vimentin, and Muc5AC (Invitrogen, UK) and counterstained with Hoechst 33258 (Invitrogen, UK), imaged using a confocal scanning laser microscope (Nikon AX/AX R with NSPARC) on day 14.

Immunohistochemistry analysis was performed on three independent biological replicates ($n = 3$) for each condition. Stained sections were imaged under identical settings, and quantitative analysis was carried out using ImageJ by calculating the CTCF to assess marker expression as per the protocol mentioned elsewhere [63].

Briefly, CTCF(per cell) = Integrated Density (cell)—(Area (cell) * Mean Gray Value (background) and the mean corrected fluorescence per cell was obtained by dividing CTCF by the corresponding cell area.

3.12 | Embedded 3D Bioprinting of Lower Respiratory Bronchial Trifurcate Structure

A 3D trifurcate bronchial tree structure was printed in our developed suspension bath with decellularized lung dECM (dLuM) using the EnvisionTEC bioplotter, Germany. A CAD model for printing the same was prepared using Fusion 360, and the .stl file was converted to G-code by Simplify 3D software into dimensions of $8.6 \times 8.47 \times 17.27$ (l x w x h). Next, the G-code file was loaded into the printing software EnvisionTEC Perfactory RP Software. A 3cc cartridge with a 330 μ m nozzle diameter was used throughout the printing process at 15°C. A further scaled-down version of the trifurcate model was designed for bioprinting with cells in 12-well plates, with dimensions $6.45 \times 6.35 \times 12.96$ (l x w x h) within the suspension bath using 3% dLuM hydrogel encapsulated with UMSCs with cell density 10^6 cells/mL using the NBIL Trivima bioprinter (Bengaluru, India). dLuM

hydrogel preparation and its rheological characterization data are provided in the Figure S15. The suspension bath used for the cell culture-based studies was prepared under sterile conditions and pH adjusted (sterile suspension bath preparation) mentioned in Section S11. Printed constructs were further evaluated with α -SMA and counterstained with Hoechst 33258, imaged using a confocal scanning laser microscope (Nikon AX/AX R with NSPARC) on day 14.

3.13 | Quantitative Real-Time Polymerase Chain Reaction (qRT-PCR)

Quantitative real-time PCR (qRT-PCR) was performed to determine the transcriptional levels of the selected genes. Total RNA was isolated from control and experimental constructs at days 1 and 14 using Ambion TRIzol Reagent (Thermo Fisher Scientific), in accordance with the manufacturer's protocol. For each condition, RNA was extracted from the entire printed construct, including all layers and regions, without spatial separation. The purity and concentration of the extracted RNA were assessed using a NanoDrop 2000 spectrophotometer by monitoring the absorbance ratio at 260/280 nm. RNA from each sample was reverse transcribed into complementary DNA (cDNA) using the PrimeScript First Strand cDNA Synthesis Kit (Takara Bio). qRT-PCR reactions were prepared in a final volume of 20 μ L containing TB Green Premix Ex Taq II (Tli RNaseH Plus; Takara Bio), 2 fold diluted cDNA, and gene-specific forward and reverse primers (final concentration: 1 pM each). Amplification was carried out on a Rotor-Gene Q real-time PCR system (QIAGEN) under the following cycling conditions: Polymerase activation at 95°C for 30 s, followed by 45 cycles of denaturation (95°C for 5 s), primer annealing (54–56°C for 30 s), and extension (72°C for 15 s). Normalization was performed using appropriate internal reference genes. For upper respiratory tract (tracheal) markers, Vimentin, Mucin, ZO-1, E-cadherin, and α -smooth muscle actin (α -SMA) were normalized to GAPDH, while the cartilage-associated genes Collagen type II (COL2A1) and Aggrecan were normalized to β -actin (ACTB). For lower respiratory tract markers, α -SMA expression was also normalized to GAPDH. Gene expression levels were normalized to the corresponding housekeeping gene Ct value. Relative mRNA abundance was calculated using the $2^{-\Delta Ct}$ method, where:

$$\Delta Ct = Ct_{\text{target}} - Ct_{\text{reference}}$$

and relative gene expression was expressed as:

$$\text{Relative expression} = 2^{-\Delta Ct}$$

To compare expression between experimental groups, the $\Delta\Delta Ct$ method was applied. The ΔCt value of each sample was normalized to the ΔCt of the control/reference group (Day 14 / Day 1), according to:

$$\Delta\Delta Ct = \Delta Ct_{D14} - \Delta Ct_{D1}$$

The fold change in expression relative to the control group was then calculated as:

$$\text{Fold change} = 2^{-\Delta\Delta Ct}$$

All measurements were performed using $n = 3$ independent samples per group.

The primer sequences of the target genes are listed in Table S3.

3.14 | Embedded Bioprinting of Tissue-Specific dECM Constructs

Different tissue-derived dECM hydrogel inks (liver, muscle, skin, cartilage, kidney, submucosa, and cornea) were prepared using pepsin digestion and standardized decellularization procedures reported elsewhere [35, 64–67]. Embedded bioprinting was performed with Envision TEC bio plotter, Germany, at optimized speed for each hydrogel ink and a 330 μm nozzle diameter in the suspension bath. Tissue-specific constructs with unique geometric or anatomical shapes were printed and thermally crosslinked at 37°C. The constructs were recovered by suspension bath leaching in PBS after printing. The Table S2, provides the printing parameters and hydrogel inks characterization. Rheological characterization of various decellularized ECM-based hydrogels fitted with the Power law model and Herschel–Bulkley model in Figure S17.

3.15 | Embedded Bioprinting Within REFRESH Bath and Control

A trifurcate model developed using CAD modeling with lung dECM hydrogel and a tongue model with skin dECM hydrogel was embedded and printed using the bio plotter (EnvisionTEC). Characteristics such as microgel formation, bath leaching time/crosslinking time, stability after printing, transparency of bath, and part quality after leaching were evaluated in conditions within the suspension bath with and without PEG mentioned in the Table S1.

3.16 | Statistical Analysis

All experiments were performed with a minimum of three independent replicates unless otherwise stated. Data was pre-processed by appropriate normalization, and outliers were assessed using standard statistical criteria. Results are presented as mean $\pm$ standard deviation (SD). Statistical significance between groups was evaluated using one-way or two-way analysis of variance (ANOVA) followed by appropriate post hoc tests or an unpaired two-tailed Student's *t*-test, as applicable. A significance level of $p < 0.05$ was considered statistically significant. ($p^* < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$; ns, not significant). Assumptions of normality and homogeneity of variance were verified prior to analysis. All statistical analyses were performed using GraphPad Prism (version 10.3.1, GraphPad Software, USA).

4 | Conclusion

In this work, we introduce REFRESH, an embedded bioprinting strategy built around a dual-function PEG–gelatin microgel suspension bath designed to address the long-standing bottleneck of printing soft, weakly crosslinked dECM hydrogels. Unlike

conventional support matrices that act only as passive mechanical scaffolds, REFRESH provides simultaneous structural confinement and macromolecular-crowding-mediated enhancement of dECM gelation, thereby stabilizing filaments during deposition while accelerating network formation. This combined mechanical and biochemical regulation markedly improves filament integrity, interlayer cohesion, and architectural fidelity, enabling the fabrication of multilayer, cell-laden constructs that closely recapitulate the cellular organization and structural complexity of native airway tissues.

REFRESH further offers several key advantages for translational biofabrication. The suspension bath exhibits robust yet tunable rheological behavior suited for high-resolution, layer-by-layer deposition; supports rapid in situ assembly of soft dECM networks, mitigating the limitations of slow thermal gelation; and maintains excellent cytocompatibility across multiple cell types, including chondrocytes, fibroblasts, epithelial cells, and stem cells. Its thermoreversible removal preserves construct geometry post-printing, while the optically transparent nature of the bath uniquely enables integration with photo-crosslinkable systems, opening opportunities for spatially controlled mechanical patterning. Moreover, because REFRESH is prepared and operated in cell-culture medium, encapsulated and surface-seeded cells remain continuously supported during crosslinking and stabilization, minimizing stress and promoting early matrix engagement. We also acknowledge the practical boundaries of the present system. The current formulations are optimized for soft dECM hydrogels and may require tuning for higher-viscosity inks or mechanically stiff materials. All constructs were cultured under static conditions, and future integration with dynamic perfusion systems will be important to drive advanced maturation. Finally, in vivo validation remains essential to establish long-term functionality, integration, and immune tolerance.

Looking ahead, the REFRESH platform is well-positioned for bioprinting other complex tubular and multilayered organs, including the esophagus, vascular and bronchial networks, and gastrointestinal tissues. The transparency-enabled photo-responsiveness further defines REFRESH as a next-generation support platform, permitting real-time polymerization and spatiotemporal control over tissue architecture. Coupled with computational modelling and patient-specific anatomical data, this system provides a clear route toward scalable, clinically relevant soft-tissue manufacturing, advancing the field toward autonomous, precision-guided bioprinting.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adhm71140-sup-0001-SuppMat.docx.

Supporting File 2: adhm71140-sup-0002-MovieS1.mp4.

Supporting File 3: adhm71140-sup-0003-MovieS2.mp4.

Supporting File 4: adhm71140-sup-0004-MovieS3.mp4.

Supporting File 5: adhm71140-sup-0005-MovieS4.mp4.

Supporting File 6: adhm71140-sup-0006-MovieS5.mp4.

Supporting File 7: adhm71140-sup-0007-MovieS6.mp4.

Supporting File 8: adhm71140-sup-0008-MovieS7.mp4.

Supporting File 9: adhm71140-sup-0009-MovieS8.mp4.